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Maximum Likelihood from Incomplete Data via the *EM* Algorithm

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SUMMARY

A broadly applicable algorithm for computing maximum likelihood estimates from incomplete data is presented at various levels of generality. Theory showing the monotone behaviour of the likelihood and convergence of the algorithm is derived. Many examples are sketched, including missing value situations, applications to grouped, censored or truncated data, finite mixture models, variance component estimation, hyperparameter estimation, iteratively reweighted least squares and factor analysis.

Keywords: MAXIMUM LIKELIHOOD; INCOMPLETE DATA; EM ALGORITHM; POSTERIOR MODE

1. INTRODUCTION

THIS paper presents a general approach to iterative computation of maximum-likelihood estimates when the observations can be viewed as incomplete data. Since each iteration of the algorithm consists of an expectation step followed by a maximization step we call it the EM algorithm. The EM process is remarkable in part because of the simplicity and generality of the associated theory, and in part because of the wide range of examples which fall under its umbrella. When the underlying complete data come from an exponential family whose maximum-likelihood estimates are easily computed, then each maximization step of an EM algorithm is likewise easily computed.

The term "incomplete data" in its general form implies the existence of two sample spaces $\mathcal{Y}$ and $\mathcal{X}$ and a many-one mapping from $\mathcal{X}$ to $\mathcal{Y}$. The observed data $\mathbf{y}$ are a realization from $\mathcal{Y}$. The corresponding $\mathbf{x}$ in $\mathcal{X}$ is not observed directly, but only indirectly through $\mathbf{y}$. More specifically, we assume there is a mapping $\mathbf{x} \rightarrow \mathbf{y}(\mathbf{x})$ from $\mathcal{X}$ to $\mathcal{Y}$, and that $\mathbf{x}$ is known only to lie in $\mathcal{X}(\mathbf{y})$, the subset of $\mathcal{X}$ determined by the equation $\mathbf{y} = \mathbf{y}(\mathbf{x})$, where $\mathbf{y}$ is the observed data. We refer to $\mathbf{x}$ as the *complete data* even though in certain examples $\mathbf{x}$ includes what are traditionally called parameters.

We postulate a family of sampling densities $f(\mathbf{x}|\boldsymbol{\phi})$ depending on parameters $\boldsymbol{\phi}$ and derive its corresponding family of sampling densities $g(\mathbf{y}|\boldsymbol{\phi})$. The complete-data specification $f(\dots|\dots)$ is related to the incomplete-data specification $g(\dots|\dots)$ by

$$g(\mathbf{y}|\boldsymbol{\phi}) = \int_{\mathcal{X}(\mathbf{y})} f(\mathbf{x}|\boldsymbol{\phi}) d\mathbf{x}. \quad (1.1)$$

The EM algorithm is directed at finding a value of $\boldsymbol{\phi}$ which maximizes $g(\mathbf{y}|\boldsymbol{\phi})$ given an observed $\mathbf{y}$, but it does so by making essential use of the associated family $f(\mathbf{x}|\boldsymbol{\phi})$. Notice that given the incomplete-data specification $g(\mathbf{y}|\boldsymbol{\phi})$, there are many possible complete-data specifications $f(\mathbf{x}|\boldsymbol{\phi})$ that will generate $g(\mathbf{y}|\boldsymbol{\phi})$. Sometimes a natural choice will be obvious, at other times there may be several different ways of defining the associated $f(\mathbf{x}|\boldsymbol{\phi})$.

Each iteration of the EM algorithm involves two steps which we call the expectation step (E-step) and the maximization step (M-step). The precise definitions of these steps, and their associated heuristic interpretations, are given in Section 2 for successively more general types of models. Here we shall present only a simple numerical example to give the flavour of the method.

Rao (1965, pp. 368–369) presents data in which 197 animals are distributed multinomially into four categories, so that the observed data consist of

$$\mathbf{y} = (y_1, y_2, y_3, y_4) = (125, 18, 20, 34).$$

A genetic model for the population specifies cell probabilities

$$(\frac{1}{2} + \frac{1}{4}\pi, \frac{1}{4}(1 - \pi), \frac{1}{4}(1 - \pi), \frac{1}{4}\pi) \text{ for some } \pi \text{ with } 0 \leq \pi \leq 1.$$

Thus

$$g(\mathbf{y}|\pi) = \frac{(y_1 + y_2 + y_3 + y_4)!}{y_1! y_2! y_3! y_4!} (\frac{1}{2} + \frac{1}{4}\pi)^{y_1} (\frac{1}{4} - \frac{1}{4}\pi)^{y_2} (\frac{1}{4} - \frac{1}{4}\pi)^{y_3} (\frac{1}{4}\pi)^{y_4}. \quad (1.2)$$

Rao uses the parameter θ where $\pi = (1 - \theta)^2$ and carries through one step of the familiar Fisher-scoring procedure for maximizing $g(\mathbf{y}|\pi)$ given the observed $\mathbf{y}$. To illustrate the EM algorithm, we represent $\mathbf{y}$ as incomplete data from a five-category multinomial population where the cell probabilities are $(\frac{1}{2}, \frac{1}{4}\pi, \frac{1}{4}(1 - \pi), \frac{1}{4}(1 - \pi), \frac{1}{4}\pi)$, the idea being to split the first of the original four categories into two categories. Thus the complete data consist of $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$ where $y_1 = x_1 + x_2$, $y_2 = x_3$, $y_3 = x_4$, $y_4 = x_5$, and the complete data specification is

$$f(\mathbf{x}|\pi) = \frac{(x_1 + x_2 + x_3 + x_4 + x_5)!}{x_1! x_2! x_3! x_4! x_5!} (\frac{1}{2})^{x_1} (\frac{1}{4}\pi)^{x_2} (\frac{1}{4} - \frac{1}{4}\pi)^{x_3} (\frac{1}{4} - \frac{1}{4}\pi)^{x_4} (\frac{1}{4}\pi)^{x_5}. \quad (1.3)$$

Note that the integral in (1.1) consists in this case of summing (1.3) over the (x_1, x_2) pairs $(0, 125), (1, 124), \dots, (125, 0)$, while simply substituting $(18, 20, 34)$ for (x_3, x_4, x_5) .

To define the EM algorithm we show how to find $\pi^{(p+1)}$ from $\pi^{(p)}$, where $\pi^{(p)}$ denotes the value of π after p iterations, for $p = 0, 1, 2, \dots$. As stated above, two steps are required. The expectation step estimates the sufficient statistics of the complete data $\mathbf{x}$, given the observed data $\mathbf{y}$. In our case, (x_3, x_4, x_5) are known to be $(18, 20, 34)$ so that the only sufficient statistics that have to be estimated are x_1 and x_2 where $x_1 + x_2 = y_1 = 125$. Estimating x_1 and x_2 using the current estimate of π leads to

$$x_1^{(p)} = 125 \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{4}\pi^{(p)}} \quad \text{and} \quad x_2^{(p)} = 125 \frac{\frac{1}{4}\pi^{(p)}}{\frac{1}{2} + \frac{1}{4}\pi^{(p)}}. \quad (1.4)$$

The maximization step then takes the estimated complete data $(x_1^{(p)}, x_2^{(p)}, 18, 20, 34)$ and estimates π by maximum likelihood as though the estimated complete data were the observed data, thus yielding

$$\pi^{(p+1)} = \frac{x_2^{(p)} + 34}{x_2^{(p)} + 34 + 18 + 20}. \quad (1.5)$$

The EM algorithm for this example is defined by cycling back and forth between (1.4) and (1.5).

Starting from an initial value of $\pi^{(0)} = 0.5$, the algorithm moved for eight steps as displayed in Table 1. By substituting $x_2^{(p)}$ from equation (1.4) into equation (1.5), and letting $\pi^* = \pi^{(p)} = \pi^{(p+1)}$ we can explicitly solve a quadratic equation for the maximum-likelihood estimate of π :

$$\pi^* = (15 + \sqrt{(53809)})/394 \div 0.6268214980.$$

The second column in Table 1 gives the deviation $\pi^{(p)} - \pi^*$, and the third column gives the ratio of successive deviations. The ratios are essentially constant for $p \geq 3$. The general theory of Section 3 implies the type of convergence displayed in this example.

The EM algorithm has been proposed many times in special circumstances. For example, Hartley (1958) gave three multinomial examples similar to our illustrative example. Other examples to be reviewed in Section 4 include methods for handling missing values in normal models, procedures appropriate for arbitrarily censored and truncated data, and estimation

TABLE 1
The EM algorithm in a simple case

p	$\pi^{(p)}$	$\pi^{(p)} - \pi^*$	$(\pi^{(p+1)} - \pi^*) \div (\pi^{(p)} - \pi^*)$
0	0.500000000	0.126821498	0.1465
1	0.608247423	0.018574075	0.1346
2	0.624321051	0.002500447	0.1330
3	0.626488879	0.000332619	0.1328
4	0.626777323	0.000044176	0.1328
5	0.626815632	0.000005866	0.1328
6	0.626820719	0.000000779	—
7	0.626821395	0.000000104	—
8	0.626821484	0.000000014	—

methods for finite mixtures of parametric families, variance components and hyperparameters in Bayesian prior distributions of parameters. In addition, the EM algorithm corresponds to certain robust estimation techniques based on iteratively reweighted least squares. We anticipate that recognition of the EM algorithm at its natural level of generality will lead to new and useful examples, possibly including the general approach to truncated data proposed in Section 4.2 and the factor-analysis algorithms proposed in Section 4.7.

Some of the theory underlying the EM algorithm was presented by Orchard and Woodbury (1972), and by Sundberg (1976), and some has remained buried in the literature of special examples, notably in Baum *et al.* (1970). After defining the algorithm in Section 2, we demonstrate in Section 3 the key results which assert that successive iterations always increase the likelihood, and that convergence implies a stationary point of the likelihood. We give sufficient conditions for convergence and also here a general description of the rate of convergence of the algorithm close to a stationary point.

Although our discussion is almost entirely within the maximum-likelihood framework, the EM technique and theory can be equally easily applied to finding the mode of the posterior distribution in a Bayesian framework. The extension required for this application appears at the ends of Sections 2 and 3.

2. DEFINITIONS OF THE EM ALGORITHM

We now define the EM algorithm, starting with cases that have strong restrictions on the complete-data specification $f(\mathbf{x}|\boldsymbol{\phi})$, then presenting more general definitions applicable when these restrictions are partially removed in two stages. Although the theory of Section 3 applies at the most general level, the simplicity of description and computational procedure, and thus the appeal and usefulness, of the EM algorithm are greater at the more restricted levels.

Suppose first that $f(\mathbf{x}|\boldsymbol{\phi})$ has the regular exponential-family form

$$f(\mathbf{x}|\boldsymbol{\phi}) = b(\mathbf{x}) \exp(\boldsymbol{\phi} \mathbf{t}(\mathbf{x})^T) / a(\boldsymbol{\phi}),$$

(2.1)

where $\boldsymbol{\phi}$ denotes a $1 \times r$ vector parameter, $\mathbf{t}(\mathbf{x})$ denotes a $1 \times r$ vector of *complete-data* sufficient statistics and the superscript T denotes matrix transpose. The term regular means here that $\boldsymbol{\phi}$ is restricted only to an r -dimensional convex set Ω such that (2.1) defines a density for all $\boldsymbol{\phi}$ in Ω . The parameterization $\boldsymbol{\phi}$ in (2.1) is thus unique up to an arbitrary non-singular $r \times r$ linear transformation, as is the corresponding choice of $\mathbf{t}(\mathbf{x})$. Such parameters are often called

natural parameters, although in familiar examples the conventional parameters are often non-linear functions of $\boldsymbol{\phi}$. For example, in binomial sampling, the conventional parameter π and the natural parameter ϕ are related by the formula $\phi = \log \pi / (1 - \pi)$. In Section 2, we adhere to the natural parameter representation for $\boldsymbol{\phi}$ when dealing with exponential families, while in Section 4 we mainly choose conventional representations. We note that in (2.1) the sample space $\mathcal{X}$ over which $f(\mathbf{x}|\boldsymbol{\phi}) > 0$ is the same for all $\boldsymbol{\phi}$ in Ω .

We now present a simple characterization of the EM algorithm which can usually be applied when (2.1) holds. Suppose that $\boldsymbol{\phi}^{(p)}$ denotes the current value of $\boldsymbol{\phi}$ after p cycles of the algorithm. The next cycle can be described in two steps, as follows:

E-step: Estimate the complete-data sufficient statistics $\mathbf{t}(\mathbf{x})$ by finding

$$\mathbf{t}^{(p)} = E(\mathbf{t}(\mathbf{x})|\mathbf{y}, \boldsymbol{\phi}^{(p)}). \quad (2.2)$$

M-step: Determine $\boldsymbol{\phi}^{(p+1)}$ as the solution of the equations

$$E(\mathbf{t}(\mathbf{x})|\boldsymbol{\phi}) = \mathbf{t}^{(p)}. \quad (2.3)$$

Equations (2.3) are the familiar form of the likelihood equations for maximum-likelihood estimation given data from a regular exponential family. That is, if we were to suppose that $\mathbf{t}^{(p)}$ represents the sufficient statistics computed from an observed $\mathbf{x}$ drawn from (2.1), then equations (2.3) usually define the maximum-likelihood estimator of $\boldsymbol{\phi}$. Note that for given $\mathbf{x}$, maximizing $\log f(\mathbf{x}|\boldsymbol{\phi}) = -\log a(\boldsymbol{\phi}) + \log b(\mathbf{x}) + \boldsymbol{\phi}^T \mathbf{t}(\mathbf{x})$ is equivalent to maximizing

$$-\log a(\boldsymbol{\phi}) + \boldsymbol{\phi}^T \mathbf{t}(\mathbf{x})$$

which depends on $\mathbf{x}$ only through $\mathbf{t}(\mathbf{x})$. Hence it is easily seen that equations (2.3) define the usual condition for maximizing $-\log a(\boldsymbol{\phi}) + \boldsymbol{\phi}^T \mathbf{t}^{(p)}$ whether or not $\mathbf{t}^{(p)}$ computed from (2.2) represents a value of $\mathbf{t}(\mathbf{x})$ associated with any $\mathbf{x}$ in $\mathcal{X}$. In the example of Section 1, the components of $\mathbf{x}$ are integer-valued, while their expectations at each step usually are not.

A difficulty with the M-step is that equations (2.3) are not always solvable for $\boldsymbol{\phi}$ in Ω . In such cases, the maximizing value of $\boldsymbol{\phi}$ lies on the boundary of Ω and a more general definition, as given below, must be used. However, if equations (2.3) can be solved for $\boldsymbol{\phi}$ in Ω , then the solution is unique due to the well-known convexity property of the log-likelihood for regular exponential families.

Before proceeding to less restricted cases, we digress to explain why repeated application of the E- and M-steps leads ultimately to the value $\boldsymbol{\phi}^*$ of $\boldsymbol{\phi}$ that maximizes

$$L(\boldsymbol{\phi}) = \log g(\mathbf{y}|\boldsymbol{\phi}), \quad (2.4)$$

where $g(\mathbf{y}|\boldsymbol{\phi})$ is defined from (1.1) and (2.1). Formal convergence properties of the EM algorithm are given in Section 3 in the general case.

First, we introduce notation for the conditional density of $\mathbf{x}$ given $\mathbf{y}$ and $\boldsymbol{\phi}$, namely,

$$k(\mathbf{x}|\mathbf{y}, \boldsymbol{\phi}) = f(\mathbf{x}|\boldsymbol{\phi})/g(\mathbf{y}|\boldsymbol{\phi}), \quad (2.5)$$

so that (2.4) can be written in the useful form

$$L(\boldsymbol{\phi}) = \log f(\mathbf{x}|\boldsymbol{\phi}) - \log k(\mathbf{x}|\mathbf{y}, \boldsymbol{\phi}). \quad (2.6)$$

For exponential families, we note that

$$k(\mathbf{x}|\mathbf{y}, \boldsymbol{\phi}) = b(\mathbf{x}) \exp(\boldsymbol{\phi}^T \mathbf{t}(\mathbf{x})) / a(\boldsymbol{\phi}|\mathbf{y}), \quad (2.7)$$

where

$$a(\boldsymbol{\phi}|\mathbf{y}) = \int_{\mathcal{X}(\mathbf{y})} b(\mathbf{x}) \exp(\boldsymbol{\phi}^T \mathbf{t}(\mathbf{x})) d\mathbf{x}. \quad (2.8)$$

Thus, we see that $f(\mathbf{x}|\boldsymbol{\phi})$ and $k(\mathbf{x}|\mathbf{y}, \boldsymbol{\phi})$ both represent exponential families with the same natural parameters $\boldsymbol{\phi}$ and the same sufficient statistics $\mathbf{t}(\mathbf{x})$, but are defined over different sample spaces $\mathcal{X}$ and $\mathcal{X}(\mathbf{y})$. We may now write (2.6) in the form

$$L(\boldsymbol{\phi}) = -\log a(\boldsymbol{\phi}) + \log a(\boldsymbol{\phi}|\mathbf{y}), \quad (2.9)$$

where the parallel to (2.8) is

$$a(\boldsymbol{\phi}) = \int_{\mathcal{X}} b(\mathbf{x}) \exp(\boldsymbol{\phi} \mathbf{t}(\mathbf{x})^T) d\mathbf{x}. \quad (2.10)$$

By parallel differentiations of (2.10) and (2.8) we obtain, denoting $\mathbf{t}(\mathbf{x})$ by $\mathbf{t}$,

$$\mathbf{D} \log a(\boldsymbol{\phi}) = (\partial/\partial \boldsymbol{\phi}) \log a(\boldsymbol{\phi}) = E(\mathbf{t}|\boldsymbol{\phi}) \quad (2.11)$$

and, similarly,

$$\mathbf{D} \log a(\boldsymbol{\phi}|\mathbf{y}) = E(\mathbf{t}|\mathbf{y}, \boldsymbol{\phi}), \quad (2.12)$$

whence

$$\mathbf{D}L(\boldsymbol{\phi}) = -E(\mathbf{t}|\boldsymbol{\phi}) + E(\mathbf{t}|\mathbf{y}, \boldsymbol{\phi}). \quad (2.13)$$

Thus the derivatives of the log-likelihood have an attractive representation as the difference of an unconditional and a conditional expectation of the sufficient statistics. Formula (2.13) is the key to understanding the E- and M-steps of the EM algorithm, for if the algorithm converges to $\boldsymbol{\phi}^*$, so that in the limit $\boldsymbol{\phi}^{(p)} = \boldsymbol{\phi}^{(p+1)} = \boldsymbol{\phi}^*$, then combining (2.2) and (2.3) leads to $E(\mathbf{t}|\boldsymbol{\phi}^*) = E(\mathbf{t}|\mathbf{y}, \boldsymbol{\phi}^*)$ or $\mathbf{D}L(\boldsymbol{\phi}) = \mathbf{0}$ at $\boldsymbol{\phi} = \boldsymbol{\phi}^*$.

The striking representation (2.13) has been noticed in special cases by many authors. Examples will be mentioned in Section 4. The general form of (2.13) was given by Sundberg (1974) who ascribed it to unpublished 1966 lecture notes of Martin-Löf. We note, parenthetically, that Sundberg went on to differentiate (2.10) and (2.8) repeatedly, obtaining

$$\text{and} \quad \left. \begin{aligned} \mathbf{D}^k a(\boldsymbol{\phi}) &= a(\boldsymbol{\phi}) E(\mathbf{t}^k|\boldsymbol{\phi}) \\ \mathbf{D}^k a(\boldsymbol{\phi}|\mathbf{y}) &= a(\boldsymbol{\phi}|\mathbf{y}) E(\mathbf{t}^k|\mathbf{y}, \boldsymbol{\phi}), \end{aligned} \right\} \quad (2.14)$$

where $\mathbf{D}^k$ denotes the k -way array of k th derivative operators and $\mathbf{t}^k$ denotes the corresponding k -way array of k th degree monomials. From (2.14), Sundberg obtained

$$\text{and} \quad \left. \begin{aligned} \mathbf{D}^k \log a(\boldsymbol{\phi}) &= \mathbf{K}^k(\mathbf{t}|\boldsymbol{\phi}) \\ \mathbf{D}^k \log a(\boldsymbol{\phi}|\mathbf{y}) &= \mathbf{K}^k(\mathbf{t}|\mathbf{y}, \boldsymbol{\phi}), \end{aligned} \right\} \quad (2.15)$$

where $\mathbf{K}^k$ denotes the k -way array of k th cumulants, so that finally he expressed

$$\mathbf{D}^k L(\boldsymbol{\phi}) = -\mathbf{K}^k(\mathbf{t}|\boldsymbol{\phi}) + \mathbf{K}^k(\mathbf{t}|\mathbf{y}, \boldsymbol{\phi}). \quad (2.16)$$

Thus, derivatives of any order of the log-likelihood can be expressed as a difference between conditional and unconditional cumulants of the sufficient statistics. In particular, when $k = 2$, formula (2.16) expressed the second-derivative matrix of the log-likelihood as a difference of covariance matrices.

We now proceed to consider more general definitions of the EM algorithm. Our second level of generality assumes that the complete-data specification is not a regular exponential family as assumed above, but a curved exponential family. In this case, the representation (2.1) can still be used, but the parameters $\boldsymbol{\phi}$ must lie in a curved submanifold Ω_0 of the r -dimensional convex region Ω . The E-step of the EM algorithm can still be defined as above, but Sundberg's formulae no longer apply directly, so we must replace the M-step by:

M-step: Determine $\boldsymbol{\phi}^{(p+1)}$ to be a value of $\boldsymbol{\phi}$ in Ω_0 which maximizes $-\log a(\boldsymbol{\phi}) + \boldsymbol{\phi} \mathbf{t}^{(p)T}$.

In other words, the M-step is now characterized as maximizing the likelihood assuming that $\mathbf{x}$ yields sufficient statistics $\mathbf{t}^{(p)}$. We remark that the above extended definition of the M-step, with Ω substituted for Ω_0 , is appropriate for those regular exponential family cases where equations (2.3) cannot be solved for ϕ in Ω .

The final level of generality omits all reference to exponential families. Here we introduce a new function

$$Q(\phi' | \phi) = E(\log f(\mathbf{x} | \phi') | \mathbf{y}, \phi), \quad (2.17)$$

which we assume to exist for all pairs (ϕ', ϕ) . In particular, we assume that $f(\mathbf{x} | \phi) > 0$ almost everywhere in $\mathcal{X}$ for all $\phi \in \Omega$. We now define the EM iteration $\phi^{(p)} \rightarrow \phi^{(p+1)}$ as follows:

E-step: Compute $Q(\phi | \phi^{(p)})$.

M-step: Choose $\phi^{(p+1)}$ to be a value of $\phi \in \Omega$ which maximizes $Q(\phi | \phi^{(p)})$.

The heuristic idea here is that we would like to choose ϕ^* to maximize $\log f(\mathbf{x} | \phi)$. Since we do not know $\log f(\mathbf{x} | \phi)$, we maximize instead its current expectation given the data $\mathbf{y}$ and the current fit $\phi^{(p)}$.

In the special case of exponential families

$$Q(\phi | \phi^{(p)}) = -\log a(\phi) + E(b(\mathbf{x}) | \mathbf{y}, \phi^{(p)}) + \phi \mathbf{t}^{(p)T},$$

so that maximizing $Q(\phi | \phi^{(p)})$ is equivalent to maximizing $-\log a(\phi) + \phi \mathbf{t}^{(p)T}$, as in the more specialized definitions of the M-step. The exponential family E-step given by (2.2) is in principle simpler than the general E-step. In the general case, $Q(\phi | \phi^{(p)})$ must be computed for all $\phi \in \Omega$, while for exponential families we need only compute the expectations of the r components of $\mathbf{t}(\mathbf{x})$.†

The EM algorithm is easily modified to produce the posterior mode of ϕ in place of the maximum likelihood estimate of ϕ . Denoting the log of the prior density by $G(\phi)$, we simply maximize $Q(\phi | \phi^{(p)}) + G(\phi)$ at the M-step of the $(p+1)$ st iteration. The general theory of Section 3 implies that $L(\phi) + G(\phi)$ is increasing at each iteration and provides an expression for the rate of convergence. In cases where $G(\phi)$ is chosen from a standard conjugate family, such as an inverse gamma prior for variance components, it commonly happens that $Q(\phi | \phi^{(p)}) + G(\phi)$ has the same functional form as $Q(\phi | \phi^{(p)})$ alone, and therefore is maximized in the same manner as $Q(\phi | \phi^{(p)})$.

3. GENERAL PROPERTIES

Some basic results applicable to the EM algorithm are collected in this section. As throughout the paper, we assume that the observable $\mathbf{y}$ is fixed and known. We conclude Section 3 with a brief review of literature on the theory of the algorithm.

In addition to previously established notation, it will be convenient to write

$$H(\phi' | \phi) = E(\log k(\mathbf{x} | \mathbf{y}, \phi') | \mathbf{y}, \phi), \quad (3.1)$$

so that, from (2.4), (2.5) and (2.17),

$$Q(\phi' | \phi) = L(\phi') + H(\phi' | \phi). \quad (3.2)$$

Lemma 1. For any pair (ϕ', ϕ) in $\Omega \times \Omega$,

$$H(\phi' | \phi) \leq H(\phi | \phi), \quad (3.3)$$

with equality if and only if $k(\mathbf{x} | \mathbf{y}, \phi') = k(\mathbf{x} | \mathbf{y}, \phi)$ almost everywhere.

Proof. Formula (3.3) is a well-known consequence of Jensen's inequality. See formulae (1e.5.6) and (1e.6.6) of Rao (1965).

† A referee has pointed out that our use of the term "algorithm" can be criticized because we do not specify the sequence of computing steps actually required to carry out a single E- or M-step. It is evident that detailed implementations vary widely in complexity and feasibility.

To define a particular instance of an iterative algorithm requires only that we list the sequence of values $\phi^{(0)} \rightarrow \phi^{(1)} \rightarrow \phi^{(2)} \rightarrow \dots$ starting from a specific $\phi^{(0)}$. In general, however, the term “iterative algorithm” means a rule applicable to any starting point, i.e. a mapping $\phi \rightarrow \mathbf{M}(\phi)$ from Ω to Ω such that each step $\phi^{(p)} \rightarrow \phi^{(p+1)}$ is defined by

$$\phi^{(p+1)} = \mathbf{M}(\phi^{(p)}). \quad (3.4)$$

Definition. An iterative algorithm with mapping $\mathbf{M}(\phi)$ is a generalized EM algorithm (a GEM algorithm) if

$$Q(\mathbf{M}(\phi) | \phi) \geq Q(\phi | \phi) \quad (3.5)$$

for every ϕ in Ω .

Note that the definitions of the EM algorithm given in Section 2 require

$$Q(\mathbf{M}(\phi) | \phi) \geq Q(\phi' | \phi) \quad (3.6)$$

for every pair (ϕ', ϕ) in $\Omega \times \Omega$, i.e. $\phi' = \mathbf{M}(\phi)$ maximizes $Q(\phi' | \phi)$.

Theorem 1. For every GEM algorithm

$$L(\mathbf{M}(\phi)) \geq L(\phi) \quad \text{for all } \phi \in \Omega, \quad (3.7)$$

where equality holds if and only if both

$$Q(\mathbf{M}(\phi) | \phi) = Q(\phi | \phi) \quad (3.8)$$

and

$$k(\mathbf{x} | \mathbf{y}, \mathbf{M}(\phi)) = k(\mathbf{x} | \mathbf{y}, \phi) \quad (3.9)$$

almost everywhere.

Proof.

$$L(\mathbf{M}(\phi)) - L(\phi) = \{Q(\mathbf{M}(\phi) | \phi) - Q(\phi | \phi)\} + \{H(\phi | \phi) - H(\mathbf{M}(\phi) | \phi)\}. \quad (3.10)$$

For every GEM algorithm, the difference in Q functions above is ≥ 0 . By Lemma 1, the difference in H functions is greater than or equal to zero with equality if and only if $k(\mathbf{x} | \mathbf{y}, \phi) = k(\mathbf{x} | \mathbf{y}, \mathbf{M}(\phi))$ almost everywhere.

Corollary 1. Suppose for some $\phi^* \in \Omega$, $L(\phi^*) \geq L(\phi)$ for all $\phi \in \Omega$. Then for every GEM algorithm,

$$(a) \quad L(\mathbf{M}(\phi^*)) = L(\phi^*),$$

$$(b) \quad Q(\mathbf{M}(\phi^*) | \phi^*) = Q(\phi^* | \phi^*)$$

and

$$(c) \quad k(\mathbf{x} | \mathbf{y}, \mathbf{M}(\phi^*)) = k(\mathbf{x} | \mathbf{y}, \phi^*) \text{ almost everywhere.}$$

Corollary 2. If for some $\phi^* \in \Omega$, $L(\phi^*) > L(\phi)$ for all $\phi \in \Omega$ such that $\phi \neq \phi^*$, then for every GEM algorithm

$$\mathbf{M}(\phi^*) = \phi^*.$$

Theorem 2. Suppose that $\phi^{(p)}$ for $p = 0, 1, 2, \dots$ is an instance of a GEM algorithm such that:

(1) the sequence $L(\phi^{(p)})$ is bounded, and

(2) $Q(\phi^{(p+1)} | \phi^{(p)}) - Q(\phi^{(p)} | \phi^{(p)}) \geq \lambda(\phi^{(p+1)} - \phi^{(p)})(\phi^{(p+1)} - \phi^{(p)})^T$ for some scalar $\lambda > 0$ and all p .

Then the sequence $\phi^{(p)}$ converges to some ϕ^* in the closure of Ω .

Proof. From assumption (1) and Theorem 1, the sequence $L(\phi^{(p)})$ converges to some $L^* < \infty$. Hence, for any $\varepsilon > 0$, there exists a $p(\varepsilon)$ such that, for all $p \geq p(\varepsilon)$ and all $r \geq 1$,

$$\sum_{j=1}^r \{L(\phi^{(p+j)}) - L(\phi^{(p+j-1)})\} = L(\phi^{(p+r)}) - L(\phi^{(p)}) < \varepsilon. \quad (3.11)$$

From Lemma 1 and (3.10), we have

$$0 \leq Q(\Phi^{(p+j)} | \Phi^{(p+j-1)}) - Q(\Phi^{(p+j-1)} | \Phi^{(p+j-1)}) \leq L(\Phi^{(p+j)}) - L(\Phi^{(p+j-1)}),$$

for $j \geq 1$, and hence from (3.11) we have

$$\sum_{j=1}^r \{Q(\Phi^{(p+j)} | \Phi^{(p+j-1)}) - Q(\Phi^{(p+j-1)} | \Phi^{(p+j-1)})\} < \varepsilon, \quad (3.12)$$

for all $p \geq p(\varepsilon)$ and all $r \geq 1$, where each term in the sum is non-negative.

Applying assumption (2) in the theorem for $p, p+1, p+2, \dots, p+r-1$ and summing, we obtain from (3.12)

$$\varepsilon > \lambda \sum_{j=1}^r (\Phi^{(p+j)} - \Phi^{(p+j-1)}) (\Phi^{(p+j)} - \Phi^{(p+j-1)})^T, \quad (3.13)$$

whence

$$\varepsilon > \lambda (\Phi^{(p+r)} - \Phi^{(p)}) (\Phi^{(p+r)} - \Phi^{(p)})^T, \quad (3.14)$$

as required to prove convergence of $\Phi^{(p)}$ to some Φ^* .

Theorem 1 implies that $L(\Phi)$ is non-decreasing on each iteration of a GEM algorithm, and is strictly increasing on any iteration such that $Q(\Phi^{(p+1)} | \Phi^{(p)}) > Q(\Phi^{(p)} | \Phi^{(p)})$. The corollaries imply that a maximum-likelihood estimate is a fixed point of a GEM algorithm. Theorem 2 provides the conditions under which an instance of a GEM algorithm converges. But these results stop short of implying convergence to a maximum-likelihood estimator. To exhibit conditions under which convergence to maximum likelihood obtains, it is natural to introduce continuity and differentiability conditions. Henceforth in this Section we assume that Ω is a region in ordinary real r -space, and we assume the existence and continuity of a sufficient number of derivatives of the functions $Q(\Phi' | \Phi)$, $L(\Phi)$, $H(\Phi' | \Phi)$ and $M(\Phi)$ to justify the Taylor-series expansions used. We also assume that differentiation and expectation operations can be interchanged.

Familiar properties of the score function are given in the following lemma, where $V[\dots | \dots]$ denotes a conditional covariance operator.

Lemma 2. For all Φ in Ω ,

$$E \left[\frac{\partial}{\partial \Phi} \log k(\mathbf{x} | \mathbf{y}, \Phi) | \mathbf{y}, \Phi \right] = \mathbf{D}^{10} H(\Phi | \Phi) = 0 \quad (3.15)$$

and

$$V \left[\frac{\partial}{\partial \Phi} \log k(\mathbf{x} | \mathbf{y}, \Phi) | \mathbf{y}, \Phi \right] = \mathbf{D}^{11} H(\Phi | \Phi) = -\mathbf{D}^{20} H(\Phi | \Phi). \quad (3.16)$$

Proof. These results follow from the definition (3.1) and by differentiating

$$\int_{\mathcal{X}(\mathbf{y})} k(\mathbf{x} | \mathbf{y}, \Phi) d\mathbf{x} = 1$$

under the integral sign.

Theorem 3. Suppose $\Phi^{(p)}$ $p = 0, 1, 2, \dots$ is an instance of a GEM algorithm such that

$$\mathbf{D}^{10} Q(\Phi^{(p+1)} | \Phi^{(p)}) = 0.$$

Then for all p , there exists a $\Phi_0^{(p+1)}$ on the line segment joining $\Phi^{(p)}$ to $\Phi^{(p+1)}$ such that

$$Q(\Phi^{(p+1)} | \Phi^{(p)}) - Q(\Phi^{(p)} | \Phi^{(p)}) = -(\Phi^{(p+1)} - \Phi^{(p)}) \mathbf{D}^{20} Q(\Phi_0^{(p+1)} | \Phi^{(p)}) (\Phi^{(p+1)} - \Phi^{(p)})^T. \quad (3.17)$$

Furthermore, if the sequence $\mathbf{D}^{20} Q(\Phi_0^{(p+1)} | \Phi^{(p)})$ is negative definite with eigenvalues bounded away from zero, and $L(\Phi^{(p)})$ is bounded, then the sequence $\Phi^{(p)}$ converges to some Φ^* in the closure of Ω .

Proof. Expand $Q(\phi | \phi^{(p)})$ about $\phi^{(p+1)}$ to obtain

$$Q(\phi | \phi^{(p)}) = Q(\phi^{(p+1)} | \phi^{(p)}) + (\phi - \phi^{(p+1)}) \mathbf{D}^{10} Q(\phi^{(p+1)} | \phi^{(p)}) \\ + (\phi - \phi^{(p+1)}) \mathbf{D}^{20} Q(\phi^{(p+1)} | \phi^{(p)}) (\phi - \phi^{(p+1)})^T$$

for some $\phi_0^{(p+1)}$ on the line segment joining ϕ and $\phi^{(p+1)}$. Let $\phi = \phi^{(p)}$ and apply the assumption of the theorem to obtain (3.17).

If the $\mathbf{D}^{20} Q(\phi_0^{(p+1)} | \phi^{(p)})$ are negative definite with eigenvalues bounded away from zero, then condition (2) of Theorem 2 is satisfied and the sequence $\phi^{(p)}$ converges to some ϕ^* in the closure of Ω since we assume $L(\phi^{(p)})$ is bounded.

Theorem 4. Suppose that $\phi^{(p)}$ $p = 0, 1, 2, \dots$ is an instance of a GEM algorithm such that

- (1) $\phi^{(p)}$ converges to ϕ^* in the closure of Ω ,
- (2) $\mathbf{D}^{10} Q(\phi^{(p+1)} | \phi^{(p)}) = \mathbf{0}$ and
- (3) $\mathbf{D}^{20} Q(\phi^{(p+1)} | \phi^{(p)})$ is negative definite with eigenvalues bounded away from zero.

Then

$$\mathbf{D}L(\phi^*) = 0, \quad (3.18)$$

$$\mathbf{D}^{20} Q(\phi^* | \phi^*) \text{ is negative definite}$$

and

$$\mathbf{D}\mathbf{M}(\phi^*) = \mathbf{D}^{20} H(\phi^* | \phi^*) [\mathbf{D}^{20} Q(\phi^* | \phi^*)]^{-1}. \quad (3.19)$$

Proof. From (3.2) we have

$$\mathbf{D}L(\phi^{(p+1)}) = -\mathbf{D}^{10} H(\phi^{(p+1)} | \phi^{(p)}) + \mathbf{D}^{10} Q(\phi^{(p+1)} | \phi^{(p)}). \quad (3.20)$$

The second term on the right-hand side of (3.20) is zero by assumption (2), while the first term is zero in the limit as $p \rightarrow \infty$ by (3.15), and hence (3.18) is established. Similarly, $\mathbf{D}^{20} Q(\phi^* | \phi^*)$ is negative definite, since it is the limit of $\mathbf{D}^{20} Q(\phi^{(p+1)} | \phi^{(p)})$ whose eigenvalues are bounded away from zero. Finally, expanding

$$\mathbf{D}^{10} Q(\phi_2 | \phi_1) = \mathbf{D}^{10} Q(\phi^* | \phi^*) + (\phi_2 - \phi^*) \mathbf{D}^{20} Q(\phi^* | \phi^*) + (\phi_1 - \phi^*) \mathbf{D}^{11} Q(\phi^* | \phi^*) + \dots, \quad (3.21)$$

and substituting $\phi_1 = \phi^{(p)}$ and $\phi_2 = \phi^{(p+1)}$, we obtain

$$\mathbf{0} = (\phi^{(p+1)} - \phi^*) \mathbf{D}^{20} Q(\phi^* | \phi^*) + (\phi^{(p)} - \phi^*) \mathbf{D}^{11} Q(\phi^* | \phi^*) + \dots \quad (3.22)$$

Since $\phi^{(p+1)} = \mathbf{M}(\phi^{(p)})$ and $\phi^* = \mathbf{M}(\phi^*)$ we obtain in the limit from (3.22)

$$\mathbf{0} = \mathbf{D}\mathbf{M}(\phi^*) \mathbf{D}^{20} Q(\phi^* | \phi^*) + \mathbf{D}^{11} Q(\phi^* | \phi^*). \quad (3.23)$$

Formula (3.19) follows from (3.2) and (3.16).

The assumptions of Theorems 3 and 4 can easily be verified in many instances where the complete-data model is a regular exponential family. Here, letting ϕ denote the natural parameters,

$$\mathbf{D}^{20} Q(\phi | \phi^{(p)}) = -V(t | \phi) \quad (3.24)$$

so that if the eigenvalues of $V(t | \phi)$ are bounded above zero on some path joining all $\phi^{(p)}$, the sequence converges. Note in this case that

$$\mathbf{D}^{20} H(\phi^* | \phi^*) = -V(t | y, \phi^*), \quad (3.25)$$

whence

$$\mathbf{D}\mathbf{M}(\phi^*) = V(t | y, \phi^*) V(t | \phi^*)^{-1}. \quad (3.26)$$

In almost all applications, the limiting ϕ^* specified in Theorem 2 will occur at a local, if not global, maximum of $L(\phi)$. An exception could occur if $\mathbf{DM}(\phi^*)$ should have eigenvalues exceeding unity. Then ϕ^* could be a saddle point of $L(\phi)$, for certain convergent $\phi^{(p)}$ leading to ϕ^* could exist which were orthogonal in the limit to the eigenvectors of $\mathbf{DM}(\phi^*)$ associated with the large eigenvalues. Note that, if ϕ were given a small random perturbation away from a saddle point ϕ^* , then the EM algorithm would diverge from the saddle point. Generally, therefore, we expect $\mathbf{D}^2 L(\phi^*)$ to be negative semidefinite, if not negative definite, in which cases the eigenvalues of $\mathbf{DM}(\phi^*)$ all lie on $[0, 1]$ or $[0, 1)$, respectively. In view of the equality, $\mathbf{D}^{20} L(\phi^*) = (\mathbf{I} - \mathbf{DM}(\phi^*)) \mathbf{D}^{20} Q(\phi^* | \phi^*)$, an eigenvalue of $\mathbf{DM}(\phi^*)$ which is unity in a neighbourhood of ϕ^* implies a ridge in $L(\phi)$ through ϕ^* .

It is easy to create examples where the parameters of the model are identifiable from the complete data, but not identifiable from the incomplete data. The factor analysis example of Section 4.7 provides such a case, where the factors are determined only up to an arbitrary orthogonal transformation by the incomplete data. In these cases, $L(\phi)$ has a ridge of local maxima including $\phi = \phi^*$. Theorem 2 can be used to prove that EM algorithms converge to particular ϕ^* in a ridge, and do not move indefinitely in a ridge.

When the eigenvalues of $\mathbf{DM}(\phi^*)$ are all less than 1, the largest such eigenvalue gives the rate of convergence of the algorithm. It is clear from (3.19) and (3.2) that the rate of convergence depends directly on the relative sizes of $\mathbf{D}^2 L(\phi^*)$ and $\mathbf{D}^{20} H(\phi^* | \phi^*)$. Note that $-\mathbf{D}^2 L(\phi^*)$ is a measure of the information in the data $\mathbf{y}$ about ϕ , while $-\mathbf{D}^{20} H(\phi^* | \phi^*)$ is an expected or Fisher information in the unobserved part of $\mathbf{x}$ about ϕ . Thus, if the information loss due to incompleteness is small, then the algorithm converges rapidly. The fraction of information loss may vary across different components of ϕ , suggesting that certain components of ϕ may approach ϕ^* rapidly using the EM algorithm, while other components may require many iterations.

We now compute the rate of convergence for the example presented in Section 1. Here the relevant quantities may be computed in a straightforward manner as

$$\mathbf{D}^{20} Q(\pi' | \pi) = -\{E(x_2 | \pi, \mathbf{y}) + y_4\} / \pi'^2 - (y_2 + y_3) / (1 - \pi')^2$$

and

$$\mathbf{D}^{20} H(\pi' | \pi) = -E(x_2 | \pi, \mathbf{y}) / \pi'^2 + y_1 / (2 + \pi')^2.$$

Substituting the value of π^* computed in Section 1 and using (3.19) we find $DM(\pi^*) \doteq 0.132778$. This value may be verified empirically via Table 1.

In some cases, it may be desirable to try to speed the convergence of the EM algorithm. One way, requiring additional storage, is to use second derivatives in order to a Newton-step. These derivatives can be approximated numerically by using data from past iterations giving the empirical rate of convergence (Aitken's acceleration process when ϕ has only one component), or by using equation (3.19), or (3.26) in the case of regular exponential families, together with an estimate of ϕ^* .

Another possible way to reduce computation when the M-step is difficult is simply to increase the Q function rather than maximize it at each iteration. A final possibility arises with missing data patterns such that factors of the likelihood have their own distinct collections of parameters (Rubin, 1974). Since the proportion of missing data is less in each factor than in the full likelihood, the EM algorithm applied to each factor will converge more rapidly than when applied to the full likelihood.

Lemma 1 and its consequence Theorem 1 were presented by Baum *et al.* (1970) in an unusual special case (see Section 4.3 below), but apparently without recognition of the broad generality of their argument. Beale and Little (1975) also made use of Jensen's inequality, but in connection with theorems about stationary points. Aspects of the theory consequent on our Lemma 2 were derived by Woodbury (1971) and Orchard and Woodbury (1972) in a general framework, but their concern was with a "principle" rather than with the EM algorithm

which they use but do not focus on directly. Convergence of the EM algorithm in special cases is discussed by Hartley and Hocking (1971) and by Sundberg (1976). We note that Hartley and Hocking must rule out ridges in $L(\phi)$ as a condition of their convergence theorem.

When finding the posterior mode, the rate of convergence is given by replacing $D^{20} Q(\phi^* | \phi^*)$ in equation (3.15) by $D^{20} Q(\phi^* | \phi^*) + D^2 G(\phi^*)$ where G is the log of the prior density of ϕ . In practice, we would expect an informative prior to decrease the amount of missing information, and hence increase the rate of convergence.

4. EXAMPLES

Subsections 4.1–4.7 display common statistical analyses where the EM algorithm either has been or can be used. In each subsection, we specify the model and sketch enough details to allow the interested reader to derive the associated E- and M-steps, but we do not study the individual algorithms in detail, or investigate the rate of convergence. The very large literature on incomplete data is selectively reviewed, to include only papers which discuss the EM algorithm or closely related theory. The range of potentially useful applications is much broader than presented here, for instance, including specialized variance components models, models with discrete or continuous latent variables, and problems of missing values in general parametric models.

4.1. *Missing Data*

Our general model involves incomplete data, and therefore includes the problem of accidental or unintended missing data. Formally, we need to assume that (a) ϕ is *a priori* independent of the parameters of the missing data process, and (b) the missing data are missing at random (Rubin, 1976). Roughly speaking, the second condition eliminates cases in which the missing values are missing because of the values that would have been observed.

We discuss here three situations which have been extensively treated in the literature, namely the multinomial model, the normal linear model and the multivariate normal model. In the first two cases, the sufficient statistics for the complete-data problem are linear in the data, so that the estimation step in the EM algorithm is equivalent to a procedure which first estimates or “fills in” the individual data points and then computes the sufficient statistics using filled-in values. In the third example, such direct filling in is not appropriate because some of the sufficient statistics are quadratic in the data values.

4.1.1. *Multinomial sampling*

The EM algorithm was explicitly introduced by Hartley (1958) as a procedure for calculating maximum likelihood estimates given a random sample of size n from a discrete population where some of the observations are assigned not to individual cells but to aggregates of cells. The numerical example in Section 1 is such a case. In a variation on the missing-data problem, Carter and Myers (1973) proposed the EM algorithm for maximum likelihood estimation from linear combinations of discrete probability functions, using linear combinations of Poisson random variables as an example. The algorithm was also recently suggested by Brown (1974) for computing the maximum-likelihood estimates of expected cell frequencies under an independence model in a two-way table with some missing cells, and by Fienberg and Chen (1976) for the special case of cross-classified data with some observations only partially classified.

We can think of the complete data as an $n \times p$ matrix $\mathbf{x}$ whose (i, j) element is unity if the i th unit belongs in the j th of p possible cells, and is zero otherwise. The i th row of $\mathbf{x}$ contains $p-1$ zeros and one unity, but if the i th unit has incomplete data, some of the indicators in the i th row of $\mathbf{x}$ are observed to be zero, while the others are missing and we know only that one of them must be unity. The E-step then assigns to the missing indicators fractions that sum to unity within each unit, the assigned values being expectations given the current estimate

of ϕ . The M-step then becomes the usual estimation of ϕ from the observed and assigned values of the indicators summed over the units.

In practice, it is convenient to collect together those units with the same pattern of missing indicators, since the filled in fractional counts will be the same for each; hence one may think of the procedure as filling in estimated counts for each of the missing cells within each group of units having the same pattern of missing data.

Hartley (1958) treated two restricted multinomial cases, namely, sampling from a Poisson population and sampling from a binomial population. In these cases, as in the example of Section 1, there is a further reduction to minimal sufficient statistics beyond the cell frequencies. Such a further reduction is not required by the EM algorithm.

4.1.2. *Normal linear model*

The EM algorithm has often been used for least-squares estimation in analysis of variance designs, or equivalently for maximum-likelihood estimation under the normal linear model with given residual variance σ^2 , whatever the value of σ^2 . A basic reference is Healy and Westmacott (1956). The key idea is that exact least-squares computations are easily performed for special design matrices which incorporate the requisite balance and orthogonality properties, while least-squares computations for unbalanced designs require the inversion of a large matrix. Thus where the lack of balance is due to missing data, it is natural to fill in the missing values with their expectations given current parameter values (E-step), then re-estimate parameters using a simple least-squares algorithm (M-step), and iterate until the estimates exhibit no important change. More generally, it may be possible to add rows to a given design matrix, which were never present in the real world, in such a way that the least-squares analysis is facilitated. The procedure provides an example of the EM algorithm. The general theory of Section 3 shows that the procedure converges to the maximum-likelihood estimators of the design parameters. The estimation of variance in normal linear models is discussed in Section 4.4.

4.1.3. *Multivariate normal sampling*

A common problem with multivariate “continuous” data is that different individuals are observed on different subsets of a complete set of variables. When the data are a sample from a multivariate normal population, there do not exist explicit closed-form expressions for the maximum-likelihood estimates of the means, variances and covariances of the normal population, except in cases discussed by Rubin (1974). Orchard and Woodbury (1972) and Beale and Little (1975) have described a cyclic algorithm for maximum-likelihood estimates, motivated by what Orchard and Woodbury call a “missing information principle”. Apart from details of specific implementation, their algorithm is an example of the EM algorithm and we believe that understanding of their method is greatly facilitated by regarding it as first estimating sufficient statistics and then using the simple complete-data algorithm on the estimated sufficient statistics to obtain parameter estimates.

We sketch here only enough details to outline the scope of the required calculations. Given a complete $n \times p$ data matrix $\mathbf{x}$ of p variables on each of n individuals, the sufficient statistics consist of p linear statistics, which are column sums of $\mathbf{x}$, and $\frac{1}{2}p(p+1)$ quadratic statistics, which are the sums of squares and sums of products corresponding to each column and pairs of columns of $\mathbf{x}$. Given a partially observed $\mathbf{x}$, it is necessary to replace the missing parts of the sums and sums of squares and products by their conditional expectations given the observed data and current fitted population parameters. Thus, for each row of $\mathbf{x}$ which contains missing values we must compute the means, mean squares and mean products of the missing values given the observed values in that row. The main computational burden is to find the parameters of the conditional multivariate normal distribution of the missing values given the observed values in that row. In practice, the rows are grouped to have a common pattern of

missing data within groups, since the required conditional normal has the same parameters within each group.

The E-step is completed by accumulating over all patterns of missing data; whereupon the M-step is immediate from the estimated first and second sample moments. The same general principles can be used to write down estimation procedures for the linear model with multivariate normal responses, where the missing data are in the response or dependent variables but not in the independent variables.

4.2. Grouping, Censoring and Truncation

Data from repeated sampling are often reported in grouped or censored form, either for convenience, when it is felt that finer reporting conveys no important information, or from necessity, when experimental conditions or measuring devices permit sample points to be trapped only within specified limits. When measuring devices fail to report even the number of sample points in certain ranges, the data are said to be truncated. Grouping and censoring clearly fall within the definition of incomplete data given in Section 1, but so also does truncation, if we regard the unknown number of missing sample points along with their values as being part of the complete data.

A general representation for this type of example postulates repeated draws of an observable z from a sample space $\mathcal{Z}$ which is partitioned into mutually exclusive and exhaustive subsets $\mathcal{Z}_0, \mathcal{Z}_1, \dots, \mathcal{Z}_t$. We suppose that (a) observations $z_{01}, z_{02}, \dots, z_{0n_0}$ are fully reported for the n_0 draws which fall in $\mathcal{Z}_0$, (b) only the numbers $n_1, n_2, \dots, n_{t-1}$ of sample draws falling in $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_{t-1}$ are reported and (c) even the number of draws falling in the truncation region $\mathcal{Z}_t$ is unknown. The observed data thus consist of $\mathbf{y} = (\mathbf{n}, \mathbf{z}_0)$, where $\mathbf{n} = (n_0, n_1, \dots, n_{t-1})$ and $\mathbf{z}_0 = (z_{01}, z_{02}, \dots, z_{0n_0})$. We denote by $n = n_0 + n_1 + \dots + n_{t-1}$ the size of the sample, *excluding* the unknown number of truncated points.

To define a family of sampling densities for the observed data $\mathbf{y} = (\mathbf{n}, \mathbf{z}_0)$, we postulate a family of densities $h(\mathbf{z}|\boldsymbol{\phi})$ over the full space $\mathcal{Z}$, and we write

$$P_i(\boldsymbol{\phi}) = \int_{\mathcal{Z}_i} h(\mathbf{z}|\boldsymbol{\phi}) d\mathbf{z} \quad \text{for } i = 0, 1, \dots, t-1,$$

and $P(\boldsymbol{\phi}) = \sum_{i=0}^{t-1} P_i(\boldsymbol{\phi})$. For given $\boldsymbol{\phi}$, we suppose that $\mathbf{n}$ has the multinomial distribution defined by n draws from t categories with probabilities $P_i(\boldsymbol{\phi})/P(\boldsymbol{\phi})$ for $i = 0, 1, \dots, t-1$, and given n_0 we treat $\mathbf{z}_0$ as a random sample of size n_0 from the density $h(\mathbf{z}|\boldsymbol{\phi})/P_0(\boldsymbol{\phi})$ over $\mathcal{Z}_0$. Thus

$$g(\mathbf{y}|\boldsymbol{\phi}) = \left(n! / \prod_{i=0}^{t-1} n_i! \right) \prod_{i=0}^{t-1} \left(\frac{P_i(\boldsymbol{\phi})}{P(\boldsymbol{\phi})} \right)^{n_i} \prod_{j=1}^{n_0} \left(\frac{h(z_{0j}|\boldsymbol{\phi})}{P_0(\boldsymbol{\phi})} \right). \quad (4.2.1)$$

A natural complete-data specification associated with (4.2.1) is to postulate $t-1$ further independent random samples, conditional on given $\mathbf{n}$ and $\boldsymbol{\phi}$, namely $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{t-1}$, where $\mathbf{z}_i = (z_{i1}, z_{i2}, \dots, z_{in_i})$ denotes n_i independent draws from the density $h(\mathbf{z}|\boldsymbol{\phi})/P_i(\boldsymbol{\phi})$ over $\mathcal{Z}_i$, for $i = 1, 2, \dots, t-1$. At this point we could declare $\mathbf{x} = (\mathbf{n}, \mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_{t-1})$, and proceed to invoke the EM machinery to maximize (4.2.1). If we did so, we would have

$$f(\mathbf{x}|\boldsymbol{\phi}) = \left(n! / \prod_{i=0}^{t-1} n_i! \right) \prod_{i=0}^{t-1} \prod_{j=1}^{n_i} \left(\frac{h(z_{ij}|\boldsymbol{\phi})}{P(\boldsymbol{\phi})} \right), \quad (4.2.2)$$

which is equivalent to regarding

$$(z_{01}, z_{02}, \dots, z_{0n_0}, z_{11}, z_{21}, \dots, z_{t-1, n_{t-1}})$$

as a random sample of size n from the truncated family $h(\mathbf{z}|\boldsymbol{\phi})/P(\boldsymbol{\phi})$ over $\mathcal{Z} - \mathcal{Z}_t$. The drawback to the use of (4.2.2) in many standard examples is that maximum likelihood estimates from a truncated family are not expressible in closed form, so that the M-step of the EM algorithm itself requires an iterative procedure.

We propose therefore a further extension of the complete data $\mathbf{x}$ to include truncated sample points. We denote by m the number of truncated sample points. Given m , we suppose that the truncated sample values $\mathbf{z}_t = (\mathbf{z}_{t1}, \mathbf{z}_{t2}, \dots, \mathbf{z}_{tm})$ are a random sample of size m from the density $h(\mathbf{z}|\boldsymbol{\phi})/(1-P(\boldsymbol{\phi}))$ over $\mathcal{Z}_t$. Finally we suppose that m has the negative-binomial density

$$l(m|n, P(\boldsymbol{\phi})) = \binom{m+n-1}{m} (1-P(\boldsymbol{\phi}))^m P(\boldsymbol{\phi})^n, \quad (4.2.3)$$

for $m = 0, 1, 2, \dots$, conditional on given $(\mathbf{n}, \mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_{t-1})$. We now have

$$\mathbf{x} = (\mathbf{n}, \mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{t-1}, m, \mathbf{z}_t)$$

whose associated sampling density given $\boldsymbol{\phi}$ is

$$f(\mathbf{x}|\boldsymbol{\phi}) = \left(n! / \prod_{i=0}^{t-1} n_i! \right) \binom{m+n-1}{m} \prod_{i=0}^t \prod_{j=1}^{n_i} h(\mathbf{z}_{ij}|\boldsymbol{\phi}). \quad (4.2.4)$$

The use of (4.2.3) can be regarded simply as a device to produce desired results, namely, (i) the $g(\mathbf{y}|\boldsymbol{\phi})$ implied by (4.2.4) is given by (4.2.1), and (ii) the complete-data likelihood implied by (4.2.4) is the same as that obtained by regarding the components of $\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_t$ as a random sample of size $n+m$ from $h(\mathbf{z}|\boldsymbol{\phi})$ on $\mathcal{Z}$.

The E-step of the EM algorithm applied to (4.2.4) requires us to calculate

$$Q(\boldsymbol{\phi}|\boldsymbol{\phi}^{(p)}) = E(\log f(\mathbf{x}|\boldsymbol{\phi})|\mathbf{y}, \boldsymbol{\phi}^{(p)}).$$

Since the combinatorial factors in (4.2.4) do not involve $\boldsymbol{\phi}$, we can as well substitute

$$\log f(\mathbf{x}|\boldsymbol{\phi}) = \sum_{i=0}^t \sum_{j=1}^{n_i} \log h(\mathbf{z}_{ij}|\boldsymbol{\phi}). \quad (4.2.5)$$

Since the $\mathbf{z}_{0i}$ values are part of the observed $\mathbf{y}$, the expectation of the $i=0$ term in (4.2.5) given $\mathbf{y}$ and $\boldsymbol{\phi}^{(p)}$ is simply

$$\sum_{j=1}^{n_0} \log h(\mathbf{z}_{0j}|\boldsymbol{\phi}).$$

For the terms $i = 1, 2, \dots, t-1$, i.e. the terms corresponding to grouping or censoring,

$$E\left(\sum_{j=1}^{n_i} \log h(\mathbf{z}_{ij}|\boldsymbol{\phi})|\mathbf{y}, \boldsymbol{\phi}^{(p)}\right) = n_i \int_{\mathcal{Z}_i} \log h(\mathbf{z}|\boldsymbol{\phi}) h(\mathbf{z}|\boldsymbol{\phi}^{(p)}) d\mathbf{z}. \quad (4.2.6)$$

For the term $i = t$ corresponding to truncation, the expression (4.2.6) still holds except that $m = n_t$ is unknown and must be replaced by its expectation under (4.2.3), so that

$$E\left(\sum_{j=1}^m \log h(\mathbf{z}_{tj}|\boldsymbol{\phi})|\mathbf{y}, \boldsymbol{\phi}^{(p)}\right) = [n/P(\boldsymbol{\phi}^{(p)})] \int_{\mathcal{Z}_t} \log h(\mathbf{z}|\boldsymbol{\phi}) h(\mathbf{z}|\boldsymbol{\phi}^{(p)}) d\mathbf{z}. \quad (4.2.7)$$

In cases where $h(\mathbf{z}|\boldsymbol{\phi})$ has exponential-family form with r sufficient statistics, the integrals in (4.2.6) and (4.2.7) need not be computed for all $\boldsymbol{\phi}$, since $\log h(\mathbf{z}|\boldsymbol{\phi})$ is linear in the r sufficient statistics. Furthermore, $Q(\boldsymbol{\phi}|\boldsymbol{\phi}^{(p)})$ can be described via estimated sufficient statistics for a (hypothetical) complete sample. Thus, the M-step of the EM algorithm reduces to ordinary maximum likelihood given the sufficient statistics from a random sample from $h(\mathbf{z}|\boldsymbol{\phi})$ over the full sample space $\mathcal{Z}$. Note that the size of the complete random sample is

$$n + E(m|n, \boldsymbol{\phi}^{(p)}) = n + n\{1 - P(\boldsymbol{\phi}^{(p)})\}/P(\boldsymbol{\phi}^{(p)}) = n/P(\boldsymbol{\phi}^{(p)}). \quad (4.2.8)$$

Two immediate extensions of the foregoing theory serve to illustrate the power and flexibility of the technique. First, the partition which defines grouping, censoring and truncation need not remain constant across sample units. An appropriate complete-data

model can be specified for the observed sample units associated with each partition and the Q -function for all units is found by adding over these collections of units. Second, independent and non-identically distributed observables, as in regression models, are easily incorporated. Both extensions can be handled simultaneously.

The familiar probit model of quantal assay illustrates the first extension. An experimental animal is assumed to live ($y = 0$) or die ($y = 1$), according as its unobserved tolerance z exceeds or fails to exceed a presented stimulus S . Thus the tolerance z is censored both above and below S . The probit model assumes an unobserved random sample $z_1, z_2, \dots, z_n$ from a normal distribution with unknown mean μ and variance σ^2 , while the observed indicators $y_1, y_2, \dots, y_n$ provide data censored at various stimulus levels $S_1, S_2, \dots, S_n$ which are supposed determined *a priori* and known. The details of the EM algorithm are straightforward and are not pursued here. Notation and relevant formulas may be found in Mantel and Greenhouse (1967) whose purpose was to remark on the special interpretation of the likelihood equations which is given in our general formula (2.13).

There is a very extensive literature on grouping, censoring and truncation, but only a few papers explicitly formulate the EM algorithm. An interesting early example is Grundy (1952) who deals with univariate normal sampling and who uses a Taylor series expansion to approximate the integrals required to handle grouping into narrow class intervals. A key paper is Blight (1970) which treats exponential families in general, and explicitly recognizes the appealing two-step interpretation of each EM iteration. Efron (1967) proposed the EM algorithm for singly censored data, and Turnbull (1974, 1976) extended Efron's approach to arbitrarily grouped, censored and truncated data.

Although Grundy and Blight formally include truncation in their discussion, they appear to be suggesting the first level of complete-data modelling, as in (4.2.2), rather than the second level, as in (4.2.4). The second-level specification was used in special cases by Hartley (1958) and Irwin (1959, 1963). Irwin ascribes the idea to McKendrick (1926). The special cases concern truncated zero-frequency counts for Poisson and negative-binomial samples. The device of assigning a negative-binomial distribution to the number of truncated sample points was not explicitly formulated by these authors, and the idea of sampling $z_{i1}, z_{i2}, \dots, z_{im}$ from the region of truncation does not arise in their special case.

4.3. Finite Mixtures

Suppose that an observable $\mathbf{y}$ is represented as n observations $\mathbf{y} = (y_1, y_2, \dots, y_n)$. Suppose further that there exists a finite set of R states, and that each y_i is associated with an unobserved state. Thus, there exists an unobserved vector $\mathbf{z} = (z_1, z_2, \dots, z_n)$, where z_i is the indicator vector of length R whose components are all zero except for one equal to unity indicating the unobserved state associated with y_i . Defining the complete data to be $\mathbf{x} = (\mathbf{y}, \mathbf{z})$, we see that the theory of Sections 2 and 3 applies for quite general specification $f(\mathbf{x}|\boldsymbol{\phi})$.

A natural way to conceptualize mixture specifications is to think first of the marginal distribution of the indicators $\mathbf{z}$, and then to specify the distribution of $\mathbf{y}$ given $\mathbf{z}$. With the exception of one concluding example, we assume throughout Section 4.3 that $z_1, z_2, \dots, z_n$ are independently and identically drawn from a density $v(\dots|\boldsymbol{\phi})$. We further assume there is a set of R densities $u(\dots|\mathbf{r}, \boldsymbol{\phi})$ for $\mathbf{r} = (1, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$ such that the y_i given z_i are conditionally independent with densities $u(\dots|\mathbf{z}_i, \boldsymbol{\phi})$. Finally, denoting

$$U(y_i|\boldsymbol{\phi}) = (\log u(y_i|(1, 0, \dots, 0), \boldsymbol{\phi}), \log u(y_i|(0, 1, \dots, 0), \boldsymbol{\phi}), \dots, \log u(y_i|(0, 0, \dots, 1), \boldsymbol{\phi})) \quad (4.3.1)$$

and

$$V(\boldsymbol{\phi}) = (\log v((1, 0, \dots, 0)|\boldsymbol{\phi}), \log v((0, 1, \dots, 0)|\boldsymbol{\phi}), \dots, \log v((0, 0, \dots, 1)|\boldsymbol{\phi})), \quad (4.3.2)$$

we can express the complete-data log-likelihood as

$$\log f(\mathbf{x}|\boldsymbol{\phi}) = \sum_{i=1}^n \mathbf{z}_i^T \mathbf{U}(\mathbf{y}_i|\boldsymbol{\phi}) + \sum_{i=1}^n \mathbf{z}_i^T \mathbf{V}(\boldsymbol{\phi}). \quad (4.3.3)$$

Since the complete-data log-likelihood is linear in the components of each $\mathbf{z}_i$, the E-step of the EM algorithm requires us to estimate the components of $\mathbf{z}_i$ given the observed $\mathbf{y}$ and the current fitted parameters. These estimated components of $\mathbf{z}_i$ are simply the current conditional probabilities that $\mathbf{y}_i$ belongs to each of the R states. In many examples, the $\boldsymbol{\phi}$ parameters of $u(\dots|\boldsymbol{\phi})$ and $v(\dots|\boldsymbol{\phi})$ are unrelated, so that the first and second terms in (4.3.3) may be maximized separately. The M-step is then equivalent to the complete-data maximization for the problem except that each observation $\mathbf{y}_i$ contributes to the log-likelihood associated with each of the R states, with weights given by the R estimated components of $\mathbf{z}_i$, and the counts in the R states are the sums of the estimated components of the $\mathbf{z}_i$.

The most widely studied examples of this formulation concern random samples from a mixture of normal distributions or other standard families. Hasselblad (1966) discussed mixtures of R normals, and subsequently Hasselblad (1969) treated more general random sampling models, giving as examples mixtures of Poissons, binomials and exponentials. Day (1969) considered mixtures of two multivariate normal populations with a common unknown covariance matrix, while Wolfe (1970) studied mixtures of binomials and mixtures of arbitrary multivariate normal distributions. Except that Wolfe (1970) referred to Hasselblad (1966), all these authors apparently worked independently. Although they did not differentiate with respect to natural exponential-family parameters, which would have produced derivatives directly in the appealing form (2.13), they all manipulated the likelihood equations into this form and recognized the simple interpretation in terms of unconditional and conditional moments. Further, they all suggested the EM algorithm. For his special case, Day (1969) noticed that the estimated marginal mean and covariance are constant across iterations, so that the implementation of the algorithm can be streamlined. All offered practical advice on various aspects of the algorithm, such as initial estimates, rates of convergence and multiple solutions to the likelihood equations. Wolfe (1970) suggested the use of Aitken's acceleration process to improve the rate of convergence. Hasselblad (1966, 1969) reported that in practice the procedure always increased the likelihood, but none of the authors proved this fact.

Two further papers in the same vein are by Hosmer (1973a, b). The first of these reported pessimistic simulation results on the small-sample mean squared error of the maximum-likelihood estimates for univariate normal mixtures, while the second studied the situation where independent samples are available from two normal populations, along with a sample from an unknown mixture of the two populations. The EM algorithm was developed for the special case of the second paper.

Haberman (1976) presented an interesting example which includes both multinomial missing values (Section 3.1.1) and finite mixtures: sampling from a multiway contingency table where the population cell frequencies are specified by a log-linear model. An especially interesting case arises when the incompleteness of the data is defined by the absence of all data on one factor. In effect, the observed data are drawn from a lower-order contingency table which is an unknown mixture of the tables corresponding to levels of the unobserved factor. These models include the clustering or latent-structure models discussed by Wolfe (1970), but permit more general and quite complex finite-mixture models, depending on the complexity of the complete-data log-linear model. Haberman showed for his type of data that each iteration of the EM algorithm increases the likelihood.

Orchard and Woodbury (1972) discussed finite-mixture problems in a non-exponential-family framework. These authors also drew attention to an early paper by Ceppellini *et al.* (1955) who developed maximum likelihood and the EM algorithm for a class of finite-mixture problems arising in genetics.

Finally, we mention another independent special derivation of the EM method for finite mixtures developed in a series of papers (Baum and Eagon, 1967; Baum *et al.*, 1970; Baum, 1972). Their model is unusual in that the n indicators $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$ are not independently and identically distributed, but rather are specified to follow a Markov chain. The complete-data likelihood given by (4.3.3) must be modified by replacing the second term by $\sum_1^n \mathbf{z}_i^T \mathbf{V}^*(\boldsymbol{\phi}) \mathbf{z}_{i-1}$ where $\mathbf{V}^*(\boldsymbol{\phi})$ is the matrix of transition probabilities and $\mathbf{z}_0$ is a known vector of initial state probabilities for the Markov chain.

4.4. Variance Components

In this section we discuss maximum-likelihood estimation of variance components in mixed-model analysis of variance. We begin with the case of all random components and then extend to the case of some fixed components.

Suppose that $\mathbf{A}$ is a fixed and known $n \times r$ “design” matrix, and that $\mathbf{y}$ is an $n \times 1$ vector of observables obtained by the linear transformation

$$\mathbf{y} = \mathbf{A}\mathbf{x} \quad (4.4.1)$$

from an unobserved $r \times 1$ vector $\mathbf{x}$. Suppose further that $\mathbf{A}$ and $\mathbf{x}$ are correspondingly partitioned into

$$\mathbf{A} = (\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{k+1}) \quad (4.4.2)$$

and

$$\mathbf{x} = (\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_{k+1}^T)^T, \quad (4.4.3)$$

where $\mathbf{A}_i$ and $\mathbf{x}_i$ have dimensions $n \times r_i$ and $r_i \times 1$ for $i = 1, 2, \dots, k+1$, and where $\sum_1^{k+1} r_i = r$. Suppose that the complete-data specification asserts that the $\mathbf{x}_i$ are independently distributed, and

$$\mathbf{x}_i \sim N(\mathbf{0}, \sigma_i^2 \mathbf{I}), \quad i = 1, \dots, k+1, \quad (4.4.4)$$

where the σ_i^2 are unknown parameters. The corresponding incomplete-data specification, implied by (1.1), asserts that $\mathbf{y}$ is normally distributed with mean vector zero and covariance matrix

$$\boldsymbol{\Sigma} = \sigma_1^2 \boldsymbol{\Sigma}_1 + \sigma_2^2 \boldsymbol{\Sigma}_2 + \dots + \sigma_{k+1}^2 \boldsymbol{\Sigma}_{k+1},$$

where the $\boldsymbol{\Sigma}_i = \mathbf{A}_i \mathbf{A}_i^T$ are fixed and known. The task is to estimate the unknown variance components $\sigma_1^2, \sigma_2^2, \dots, \sigma_{k+1}^2$.

As described, the model is a natural candidate for estimation by the EM algorithm. In practice, however, the framework usually arises in the context of linear models where the relevance of incomplete-data concepts is at first sight remote. Suppose that $\mathbf{A}_{k+1} = \mathbf{I}$ and that we rewrite (4.4.1) in the form

$$\mathbf{y} = \sum_{i=1}^k \mathbf{A}_i \mathbf{x}_i + \mathbf{x}_{k+1}. \quad (4.4.5)$$

Then we may interpret $\mathbf{y}$ as a response vector from a linear model where $(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k)$ represents a partition of the design matrix, $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k)$ represents a partition of the vector of regression coefficients and $\mathbf{x}_{k+1}$ represents the vector of discrepancies of $\mathbf{y}$ from linear behaviour. The normal linear model assumes that the components of $\mathbf{x}_{k+1}$ are independent $N(0, \sigma^2)$ distributed, as we have assumed with $\sigma^2 = \sigma_{k+1}^2$. Making the $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ also normally distributed, as we did above, converts the model from a fixed effects model to a random effects model.

When the model is viewed as an exponential family of the form (2.1), the sufficient statistics are

$$\mathbf{t}(\mathbf{x}) = (\mathbf{x}_1^T \mathbf{x}_1, \mathbf{x}_2^T \mathbf{x}_2, \dots, \mathbf{x}_{k+1}^T \mathbf{x}_{k+1}). \quad (4.4.6)$$

The E-step requires us to calculate the conditional expectations of $t_i = \mathbf{x}_i^T \mathbf{x}_i$ given $\mathbf{y}$ and the current $\sigma_i^{(p)2}$, for $i = 1, 2, \dots, k+1$. Standard methods can be used to compute the mean $\boldsymbol{\mu}_i^{(p)}$ and covariance $\boldsymbol{\Sigma}_i^{(p)}$ of the conditional normal distributions of the $\mathbf{x}_i$, given $\mathbf{y}$ and the current parameters, from the joint normal distribution specified by (4.4.1)–(4.4.4) with $\sigma_i^{(p)2}$ in place of σ_i^2 . Then the conditional expectations of $\mathbf{x}_i^T \mathbf{x}_i$ are

$$t_i^{(p)} = \boldsymbol{\mu}_i^{(p)} \boldsymbol{\mu}_i^{(p)T} + \text{tr } \boldsymbol{\Sigma}_i^{(p)}. \quad (4.4.7)$$

The M-step of the EM algorithm is then trivial since the maximum-likelihood estimators of the σ_i^2 given $t_i^{(p)}$ are simply

$$\sigma_i^{(p+1)2} = t_i^{(p)} / r_i \quad \text{for } i = 1, 2, \dots, k+1. \quad (4.4.8)$$

Random effects models can be viewed as a special subclass of mixed models where the fixed effects are absent. To define a general mixed model, suppose that $\mathbf{x}_1$ in (4.4.3) defines unknown fixed parameters, while $\mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_{k+1}$ are randomly distributed as above. Then the observed data $\mathbf{y}$ have a normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, where

$$\boldsymbol{\mu} = \mathbf{A}_1 \mathbf{x}_1 \quad \text{and} \quad \boldsymbol{\Sigma} = \sum_{i=2}^{k+1} \sigma_i^2 \boldsymbol{\Sigma}_i. \quad (4.4.9)$$

Maximum likelihood estimates of $\mathbf{x}_1, \sigma_2^2, \dots, \sigma_{k+1}^2$ can be obtained by the EM method where $(\mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_{k+1})$ are regarded as missing. We do not pursue the details, but we note that the iterative algorithm presented by Hartley and Rao (1967) for the mixed model is essentially the EM algorithm.

An alternative approach to the mixed model is to use a pure random effects analysis except that σ_1 is fixed at ∞ . Again the EM algorithm can be used. It can be shown that the estimates of $\sigma_2, \sigma_3, \dots, \sigma_{k+1}$ found in this way are identical to those described by Patterson and Thompson (1971), Corbeil and Searle (1976) and Harville (1977) under the label REML, or “restricted” maximum likelihood.

4.5. Hyperparameter Estimation

Suppose that a vector of observables, $\mathbf{y}$, has a statistical specification given by a family of densities $l(\mathbf{y}|\boldsymbol{\theta})$ while the parameters $\boldsymbol{\theta}$ themselves have a specification given by the family of densities $h(\boldsymbol{\theta}|\boldsymbol{\phi})$ depending on another level of parameters $\boldsymbol{\phi}$ called the hyperparameters. Typically, the number of components in $\boldsymbol{\phi}$ is substantially less than the number of components in $\boldsymbol{\theta}$. Such a model fits naturally into our incomplete data formulation when we take $\mathbf{x} = (\mathbf{y}, \boldsymbol{\theta})$. Indeed, the random effect model studied in (4.4.5) is an example, where we take $\boldsymbol{\theta}$ to be $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k, \sigma^2)$ and $\boldsymbol{\phi}$ to be $(\sigma_1^2, \sigma_2^2, \dots, \sigma_k^2)$.

Bayesian models provide a large fertile area for the development of further examples. Traditional Bayesian inference requires a specific prior density for $\boldsymbol{\theta}$, say $h(\boldsymbol{\theta}|\boldsymbol{\phi})$ for a specific $\boldsymbol{\phi}$. When $h(\boldsymbol{\theta}|\boldsymbol{\phi})$ is regarded as a family of prior densities, a fully Bayesian approach requires a “hyperprior” density for $\boldsymbol{\phi}$. Section 3 pointed out that the EM algorithm can be used to find the posterior mode for such problems. An *ad hoc* simplification of the fully Bayesian approach involves inferences about $\boldsymbol{\theta}$ being drawn using the prior density $h(\boldsymbol{\theta}|\boldsymbol{\phi})$ with $\boldsymbol{\phi}$ replaced by a point estimate $\hat{\boldsymbol{\phi}}$. These procedures are often called empirical Bayes’ procedures. Many examples and a discussion of their properties may be found in Maritz (1964). Other examples involving the use of point estimates of $\boldsymbol{\phi}$ are found in Mosteller and Wallace (1965), Good (1967) and Efron and Morris (1975).

A straightforward application of the EM algorithm computes the maximum-likelihood estimate of $\boldsymbol{\phi}$ from the marginal density of the data $g(\mathbf{y}|\boldsymbol{\phi})$, here defined as

$$g(\mathbf{y}|\boldsymbol{\phi}) = \int_{\boldsymbol{\theta}} l(\mathbf{y}|\boldsymbol{\theta}) h(\boldsymbol{\theta}|\boldsymbol{\phi}) d\boldsymbol{\theta}$$

for $\theta \in \Theta$. The E-step tells us to estimate $\log f(\mathbf{x}|\phi) = \log l(\mathbf{y}|\theta) + \log h(\theta|\phi)$ by its conditional expectation given $\mathbf{y}$ and $\phi = \phi^{(p)}$. For the M-step, we maximize this expectation over ϕ . When the densities $h(\theta|\phi)$ form an exponential family with sufficient statistics $\mathbf{t}(\theta)$, then $f(\mathbf{x}|\phi)$ is again an exponential family with sufficient statistics $\mathbf{t}(\theta)$, regardless of the form of $l(\mathbf{y}|\theta)$, whence the two steps of the EM algorithm reduce to (2.2) and (2.3).

It is interesting that the EM algorithm appears in the Bayesian literature in Good (1956), who apparently found it appealing on intuitive grounds but did not recognize the connection with maximum likelihood. He later (Good, 1965) discussed estimation of hyperparameters by maximum likelihood for the multinomial-Dirichlet model, but without using EM.

4.6. Iteratively Reweighted Least Squares

For certain models, the EM algorithm becomes iteratively reweighted least squares. Specifically, let $\mathbf{y} = (y_1, \dots, y_n)$ be a random sample from a population such that $(y_i - \mu)\sqrt{q_i}/\sigma$ has a $N(0, 1)$ distribution conditional on q_i , and $\mathbf{q} = (q_1, \dots, q_n)$ is an independently, identically distributed sample from the density $h(q_i)$ on $q_i \geq 0$. When q_i is unobserved, the marginal density of y_i has the form given by (1.1) and we may apply the EM algorithm to estimate μ and σ^2 . As an example, when $h(q_i)$ defines a χ_r^2 distribution, then the marginal distribution of y_i is a linearly transformed t with r degrees of freedom. Other examples of “normal/independent” densities, such as the “normal/uniform” or the contaminated normal distribution may be found in Chapter 4 of Andrews *et al.* (1972).

First suppose $h(q_i)$ is free of unknown parameters. Then the density of $\mathbf{x} = (\mathbf{y}, \mathbf{q})$ forms an exponential family with sufficient statistics $\sum y_i^2 q_i$, $\sum y_i q_i$ and $\sum q_i$. When $\mathbf{q}$ is observed the maximum likelihood estimates of μ and σ^2 are obtained from a sample of size n by simple weighted least squares:

$$\left. \begin{aligned} \hat{\mu} &= \frac{\sum_{i=1}^n y_i q_i}{\sum_{i=1}^n q_i}, \\ \hat{\sigma}^2 &= \frac{\sum_{i=1}^n (y_i - \hat{\mu})^2 q_i}{n}. \end{aligned} \right\} \quad (4.6.1)$$

When $\mathbf{q}$ is not observed, we may apply the EM algorithm:

E-step: Estimate $(\sum y_i^2 q_i, \sum y_i q_i, \sum q_i)$ by its expectation given $\mathbf{y}$, $\mu^{(p)}$ and $\sigma^{(p)2}$.

M-step: Use the estimated sufficient statistics to compute $\mu^{(p+1)}$ and $\sigma^{(p+1)2}$.

These steps may be expressed simply in terms of equations (4.6.1), indexing the left-hand sides by $(p+1)$, and substituting

$$w_i = E(q_i | y_i, \hat{\mu}^{(p)}, \hat{\sigma}^{(p)2}) \quad (4.6.2)$$

for q_i on the right-hand side. The effect of not observing $\mathbf{q}$ is to change the simple weighted least-squares equations to iteratively reweighted least-squares equations.

We remark that w_i is easy to find for some densities $h(q_i)$. For example, if

$$h(q_i) = (\beta^\alpha / \Gamma(\alpha)) q_i^{\alpha-1} \exp(-\beta q_i) \quad (4.6.3)$$

for $\alpha, \beta, q_i > 0$, then $h(q_i | y_i, \mu^{(p)}, \sigma^{(p)2})$ has the same gamma form with α and β replaced by $\alpha^* = \alpha + \frac{1}{2}$ and $\beta_i^* = \beta + \frac{1}{2}(y_i - \mu^{(p)})^2 / \sigma^{(p)2}$, whence

$$w_i = \alpha^* / \beta_i^*.$$

To obtain a contaminated normal, we may set

$$h(q_i) = \begin{cases} \alpha_1 & \text{if } q_i = k_1, \\ \alpha_2 & \text{if } q_i = k_2, \\ 0 & \text{otherwise,} \end{cases}$$

where $\alpha_i > 0$, $\alpha_1 + \alpha_2 = 1$. Then

$$w_i = \frac{\sum_{j=1}^2 k_j^{\frac{3}{2}} \alpha_j \exp(-z_{ij})}{\sum_{j=1}^2 k_j^{\frac{1}{2}} \alpha_j \exp(-z_{ij})},$$

where

$$z_{ij} = (y_i - \mu^{(p)})^2 k_j / \{2\sigma^{(p)2}\}.$$

If $h(q_i)$ is uniform on (a, b) , then $h(q_i | y_i, \mu^{(p)}, \sigma^{(p)})$ is simply proportional to the density of y_i given q_i , $\mu^{(p)}$ and $\sigma^{(p)}$. Since this conditional density of y_i is $N(\mu^{(p)}, \sigma^{(p)2}/q_i)$, $h(q_i | y_i, \mu^{(p)}, \sigma^{(p)})$ has the form given in (4.6.3) with $a < q_i < b$, $\alpha = 3$ and $\beta = (y_i - \mu^{(p)})^2 / \{2\sigma^{(p)2}\}$. In this last example, computation of w_i requires evaluation of incomplete gamma functions.

We may also allow $h(q_i)$ to depend on unknown parameters, say λ , which must be estimated with μ and σ^2 . For example, when $h(q_i)$ is χ_r^2 with unknown r , then r must be estimated. If λ is distinct from μ and σ^2 , then the complete-data log-likelihood, and hence

$$Q(\mu, \sigma^2, \lambda | \mu^{(p)}, \sigma^{(p)2}, \lambda^{(p)})$$

is the sum of two pieces, one depending only on (μ, σ^2) , the other depending only on λ . Implementing the EM algorithm by maximizing $Q(\dots | \dots)$ again leads to iteratively reweighted least squares for $\mu^{(p+1)}$ and $\sigma^{(p+1)2}$, with additional equations for $\lambda^{(p+1)}$.

4.7. Factor Analysis

In our final class of examples, interest focuses on the dependence of p observed variables on $q < p$ unobserved “latent” variables or “factors”. When both sets of variables are continuous and the observed variables are assumed to have a linear regression on the factors, the model is commonly called factor analysis. Our discussion using the EM algorithm applies when the variables are normally distributed.

More specifically, let $\mathbf{y}$ be the $n \times p$ observed data matrix and $\mathbf{z}$ be the $n \times q$ unobserved factor-score matrix. Then $\mathbf{x} = (\mathbf{y}, \mathbf{z})$, where the rows of $\mathbf{x}$ are independently and identically distributed. The marginal distribution of each row of $\mathbf{z}$ is normal with mean $(0, \dots, 0)$, variance $(1, \dots, 1)$ and correlation $\mathbf{R}$. The conditional distribution of the i th row of $\mathbf{y}$ given $\mathbf{z}$ is normal with mean $\alpha + \beta \mathbf{z}_i$ and residual covariance $\tau^2 = \text{diag}(\tau_1^2, \dots, \tau_p^2)$, where $\mathbf{z}_i$ is the i th row of $\mathbf{z}$. Note that given the factors the variables are independent. The parameters ϕ thus consist of α , β and τ^2 . The regression coefficient matrix β is commonly called the factor-loading matrix and the residual variances τ^2 are commonly called the uniquenesses.

Two cases are defined by further restrictions on β and/or $\mathbf{R}$. In the first case, β is unrestricted and $\mathbf{R} = \mathbf{I}$. In the second case, restrictions are placed on β (*a priori* zeroes), and the requirement that $\mathbf{R} = \mathbf{I}$ is possibly relaxed so that some of the correlations among the factors are to be estimated. See Jöreskog (1969) for examples and discussion of these models. It is sometimes desirable to place a prior distribution on the uniquenesses to avoid the occurrence of zero estimates (Martin and McDonald, 1975).

If the factors were observed, the computation of the maximum-likelihood estimates of ϕ would follow from the usual least-squares computations based on the sums, sums of squares, and sum of cross-products of $\mathbf{x}$. Let $(\bar{\mathbf{y}}, \bar{\mathbf{z}})$ be the sample mean vector and

$$\begin{bmatrix} C_{yy} & C_{yz} \\ C_{zy} & C_{zz} \end{bmatrix}$$

be the sample cross-products matrix of $\mathbf{x}$. Then the maximum-likelihood estimate of α is simply $\bar{\mathbf{y}}$ while the maximum-likelihood estimates of the factor loadings and uniqueness for the j th variable follow from the regression of that variable on the factors. Note that the calculations of these parameters may involve different sets of factors for different observed variables reflecting the *a priori* zeros in β . The matrix $\mathbf{R}$ is estimated from C_{zz} (and $\bar{\mathbf{z}}$); if

restrictions are placed on $\mathbf{R}$, special complete-data maximum-likelihood techniques may have to be used (Dempster, 1972). We have thus described the M-step of the algorithm, namely, the computation of the maximum-likelihood estimate of ϕ from complete data. The algorithm can be easily adapted to obtain the posterior mode when prior distributions are assigned to the uniqueness.

The E-step of the algorithm requires us to calculate the expected value of C_{zz} and C_{zy} given the current estimated ϕ ($\bar{z}$ is always estimated as $\mathbf{0}$). This computation is again a standard least-squares computation: we estimate the regression coefficients of the factors on the variables assuming the current estimated ϕ found from the M-step.

Thus the resultant EM-algorithm consists of “back and forth” least-squares calculations on the cross-products matrix of all variables (with the M-step supplemented in cases of special restrictions on $\mathbf{R}$). Apparently, the method has not been previously proposed, even though it is quite straightforward and can handle many cases using only familiar computations.

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DISCUSSION ON THE PAPER BY PROFESSOR DEMPSTER, PROFESSOR LAIRD AND DR RUBIN

E. M. L. BEALE (Scicon Computer Services Ltd and Scientific Control Systems Ltd): It gives me great pleasure to open the discussion of this lucid and scholarly paper on an important topic, and to thank all three authors for crossing the Atlantic to present it to us. The topic is in many ways a deceptive one, so it is hardly surprising that earlier authors have seen only parts of it. I therefore thought it might be useful to relate the development of Dr Little's and my understanding of the subject. We were studying multiple linear regression with missing values, and we developed an iterative algorithm that worked well in simulation experiments. We justified it on the grounds that it produced consistent estimates, but we were not clear about its relation to maximum likelihood. And when we saw Orchard and Woodbury's paper we had difficulty in understanding it. You must make allowance for the fact that at the time Rod Little was a young Ph.D. student, with a mere one-day-a-week visiting professor for a supervisor. Our difficulty was essentially a

chicken-and-egg problem. It was clear that if we knew the values of the parameters, then we could find unbiased estimates for the sufficient statistics from the underlying complete data, and hence find maximum-likelihood estimates for the parameters. But was this a fundamentally circular argument?

We eventually understood that it was not a circular argument, but a transformation from an assumed parameter-vector to another parameter-vector that maximized the conditional expected likelihood. And we also understood Orchard and Woodbury's Missing Information Principle as proclaiming that the maximum-likelihood solution is a fixed point of this transformation. Orchard and Woodbury also used the Missing Information Algorithm—the present authors' EM algorithm—as a way of finding this fixed point. But they also observed—as have the present authors—that the algorithm is not always an effective way of exploiting the principle.

These remarks explain why I am less enthusiastic about the authors' new terminology than I am about the technical content of their paper. This applies particularly to the general form of the algorithm expounded after equation (2.17), where the division of the method into separate E- and M-steps is quite impractical.

The fact remains that the authors have added considerably to our understanding of the algorithm, as well as drawing attention to its wide applicability. Theorem 1, proving that it always increases the likelihood, is very comforting. If we assume that the set of parameter values giving a higher value of the likelihood is bounded, then we can deduce that the sequence $\phi^{(p)}$ has at least one limit point, and any such limit point must be either a maximum or a stationary value of $L(\phi)$ (provided that $\phi^{(p+1)}$ is chosen so that $Q(\phi | \phi^{(p)})$ is significantly larger for $\phi = \phi^{(p+1)}$ than for $\phi = \phi^{(p)}$ whenever $\phi = \phi^{(p)}$ is a significant distance from a maximum or stationary value of the function).

It would be interesting to know more about the circumstances in which this limit point was guaranteed to be a global maximum of $L(\phi)$. The fact that the global maximum may not be unique indicates that this is a difficult question.

Practical people might well argue that this question is of more mathematical than statistical interest. So let me now say a few words about something practical we have recently done about missing values in multiple regression. The obvious practical approach is to guess at suitable values for the missing quantities and then to analyse the data as if they were complete. I understand that a more respectable name for these guessed values is "imputed values" if the guessing is done objectively. Now the natural approach is to replace each missing quantity by its mean value conditional on the values of all known variables. And these mean values can be found iteratively by the Missing Information Algorithm. But the Missing Information Principle shows that this may be unsatisfactory if the sufficient statistics are the first and second moments of the variables, since there is a negative bias in the resulting estimates of the variances. We overcome this difficulty by replacing missing values by two or more alternative sets of imputed values. Specifically, if an observation contains m missing values, we replace it with $(m+1)$ equally weighted fractional observations containing imputed values that match the first and second moments. So we use the Missing Information Algorithm to impute appropriate sets of values for the missing quantities, and then analyse the data using a standard multiple regression program. This simplifies the data-processing aspects of the work and, in particular, the task of residual analysis. It also allows us to transform the data if we are prepared to give up strict maximum likelihood. We know there are negative biases in the conventional standard errors calculated as if incomplete data were complete, so we have the option to use Beale and Little's approximation instead.

In conclusion, I am happy to propose a vote of thanks for this paper, in spite of its title.

Dr J. A. NELDER (Rothamsted Experimental Station): There are times when it seems that our subject is becoming more and more fragmented with people quarrying vigorously in smaller and smaller corners and becoming more and more isolated from each other in consequence. It is a particular pleasure, therefore, to be able to speak to a paper which seeks, and achieves, a synthesis by exhibiting diverse procedures as special cases of a general one.

The EM algorithm is particularly simple for exponential families with a canonical form for the parameters, and not especially complicated for other parametrizations. However, the authors push the algorithm towards complete generality in equation (2.17) and the succeeding definitions. In this most general form the procedure looks less attractive, because the E-step now requires the evaluation of a multi-dimensional function of continuous variables, possibly with infinite ranges, in contrast to the computation of a finite set of quantities for the simpler cases. Such a function can of course only be approximated by a set of values on a finite grid in any actual algorithm, and the practical

problems involved in the selection of such a grid are not trivial. It is probably not accidental that in their examples the authors do not need this level of generality, and indeed they demonstrate that the more specific form has a very wide scope.

There is, as the authors recognize, often a considerable gap between the establishment of a theoretical result, that a function converges to a limit, and the construction of an efficient algorithm to evaluate that limit. It was Jeffreys, I think, who pointed out that a convergent series was fine in theory, but could be hopelessly slow to converge in practice, while a divergent series, if asymptotic, might provide a very good algorithm but could not give the function value to arbitrary accuracy. Experience with this algorithm suggests to me that a good deal of work will be justified to improve convergence in common and important cases. I give two examples from practical experience.

The first is the missing value problem for the Normal linear model, discussed in Section 4.1.2. It is well known that for balanced cross-classified data with an additive model and one missing value, convergence in one cycle can be achieved by multiplying the calculated adjustment to the starting value, as given by the EM algorithm, by the factor N/ν where N is the total number of observations and ν the number of d.f. for error with the data complete. Preece (1971) has investigated to what extent this result can be generalized. When there are several missing values, we noticed that the EM algorithm was often rather slow, and Preece devised an adjusted algorithm using a suitable "stretching" factor for as long as the likelihood continued to increase (stretching may cause divergence occasionally). Preece's algorithm is a substantial improvement on the plain EM algorithm.

The second example concerns latent-structure analysis. I investigated a 4×4 table of counts which was supposed to be the margin of a hypothetical $4 \times 4 \times 2$ table in which the two slices could be fitted by a multiplicative model, but with different effects in the two slices. The ratio of the totals in the two slices is also assumed unknown. The procedure described by Goodman (1974) is an example of the EM algorithm. Given a working set of parameters for the $4 \times 4 \times 2$ table, the complete data can be estimated by adjusting the fitted values from the model to add to the known 4×4 margin. The model is then refitted to these adjusted data. Each cycle increased the likelihood, but often painfully slowly. However, I noticed that if μ_1 and μ_2 were two consecutive sets of estimated complete data for the $4 \times 4 \times 2$ table then although μ_2 was an improvement on μ_1 , exploration in the direction $\ln \mu_2 - \ln \mu_1$ was often very successful. In fact it appeared that the step $\ln \mu_2 - \ln \mu_1$, though in the right direction, could be too small in size by a factor as large as 10. Here again a simple stretching addition to the basic algorithm brought substantial improvement.

I found the authors' description of factor analysis in Section 4.7 most enlightening, and I now realize that factor analysis and latent-structure analysis are both instances of generalized linear models with unknown covariates. For factor analysis we have Normal errors and quantitative covariates, and for latent-structure analysis Poisson errors and qualitative covariates. However, I have been most frustrated by my inability to get their factor-analysis algorithm to work, and I hope they will give a full algebraic specification in their reply.

I hope it is obvious that I have found this paper intellectually stimulating, and I am very glad to second the vote of thanks.

The vote of thanks was passed by acclamation.

Professor CEDRIC A. B. SMITH (University College London): The iterative algorithm presented by the authors has a number of virtues not explicitly mentioned. For example, at least for (possibly truncated and mixed) multinomial data their Table 1 readily leads to the standard error. Denote the ratio in the last column by λ ($= 0.1328$, after a few iterations). The total number of individuals in the denominator of (1.5) can be estimated as

$$x_2^* + 34 + 18 + 20 = 125 \frac{\frac{1}{2}\pi^*}{\frac{1}{2} + \frac{1}{2}\pi^*} + 34 + 18 + 20 = n^* \quad (\text{say}) \quad (= 101.83)$$

The standard error of π^* is then

$$\text{s.e.}(\pi^*) = \sqrt{\frac{\pi^*(1-\pi^*)}{n^*(1-\lambda)}} \quad (= 0.051).$$

This is the ordinary binomial standard error inflated by an extra factor $(1-\lambda)$ in the denominator.

There are other features adding generality and flexibility to the algorithm. Thus Aitken's acceleration process to speed up the convergence can be thought of geometrically. For each value $\pi^{(p)}$, (1.4) and (1.5) give a corresponding "improved" value $\pi^{(p+1)}$. The points $(x, y) = (\pi^{(p)}, \pi^{(p+1)})$ lie on a curve C intersecting the line $(x = y)$ at (π^*, π^*) . A straight line through two points $(\pi^{(p)}, \pi^{(p+1)})$ intersects $x = y$ near the final point (π^*, π^*) . A parabola through three points is better.

However, in particular cases there are methods more powerful than EM. The maximum-likelihood equation for the model of (1.2) is

$$\frac{125}{2+\pi} - \frac{18}{1-\pi} - \frac{20}{1-\pi} + \frac{34}{\pi} = 0.$$

As M. C. K. Tweedie suggested (1945, see Smith, 1969, pp. 421-423) by replacing each term by its reciprocal we get the easily solvable linear equation

$$\frac{2+\pi}{125} - \frac{1-\pi}{18} - \frac{1-\pi}{20} + \frac{\pi}{34} = 0.$$

The solution is $0.089555/0.142967 = 0.6264$, differing only trivially from $\pi^* = 0.6268$. Its standard error is $(\frac{1}{4} \times 197 \times \sqrt{0.147967})^{-1} = 0.0515$.

Dr R. J. A. LITTLE (ISI World Fertility Survey): The authors have produced an excellent review paper demonstrating the power and elegance of this algorithm, and have contributed usefully to its theory. To place the algorithm in context, let us compare it with the most common alternative iterative technique for maximum-likelihood estimation, the method of scoring.

Advantages of the method of scoring are a quadratic rather than linear convergence rate, and the provision of an estimated asymptotic covariance matrix of the maximum-likelihood estimates, from the final inversion of the (expected) information matrix at the Newton step. In contrast, the EM algorithm does not provide estimates of standard error, since calculation and inversion of the information matrix are avoided. In another sense this is an advantage, since in multiparameter examples such as the incomplete multivariate normal sample or factor analysis, the information matrix can have high dimension and inverting it may be costly.

Other advantages of the EM approach are (a) because it is stupid, it is safe, (b) it is easy to program, and often allows simple adaptation of complete data methods, and (c) it provides fitted values for missing data.

A useful illustration of these merits lies in the linear model with unbalanced design matrix. The question of speeding the algorithm has been raised in this context, so let me suggest that considerable gains can be obtained by judicious choice of the form of the hypothetical complete data $\mathbf{x}$. For example, Chapter 16 of Snedecor and Cochran (1967) gives a 3×3 table of means $\{\bar{y}_{ij}\}$ for an unbalanced two-way analysis of variance, with cell counts $\{n_{ij}\} = (36, 67, 49, 31, 60, 49, 58, 87, 80)$. A naive application of the Healy-Westmacott procedure would be to invent missing observations so that $\mathbf{x}$ is a balanced table with 87 observations per cell. However if $\mathbf{x}$ had cell counts n'_{ij} such that $n'_{ij} = a_i b_j$, then a simple analysis is possible on the cell totals, as Snedecor and Cochran point out. Hence we seek $\mathbf{x}$ with cell counts $n'_{ij} = n_{ij} + m_{ij}$ and $n'_{ij} = a_i b_j$. Fractional values of m_{ij} are allowed. One choice of $\{n'_{ij}\}$, not optimal, is $\{n'_{ij}\} = (39.1, 67, 55.7, 35.1, 60, 49.9, 58, 99.3, 82.6)$. We have drastically reduced the amount of missing data and thus, by the authors' Theorem 2, increased the rate of convergence. Here the E-step consists in calculating from current estimates $\mu_{ij}^{(r)}$ of the cell means cell totals $n_{ij} \bar{y}_{ij} + m_{ij} \mu_{ij}^{(r)}$. The M-step is the standard analysis with these totals, and cell sizes n'_{ij} . Inventing data is an unusual practice to recommend to this Society, but here it does quite nicely!

Finally, I should like to add to the authors' impressive list of applications discriminant analysis and time series analysis. McClachlan (1975) has given a procedure for using unclassified observations in discriminant analysis to help estimate the discriminant function. This method can be improved by treating the unclassified data as incomplete, in that the variable indicating the group is missing, and applying the EM algorithm. However, in practice I have found that gains over standard procedures are negligible. Perhaps a more promising application lies in estimation of autoregressive models for time series with missing values, as suggested in Little (1974).

Mr T. J. ORCHARD (Office of Population Censuses and Surveys): I should like to congratulate the speakers on the presentation of an interesting and valuable paper. To some extent they have merely restated what has been presented in the papers referred to, but the material has been presented in such a mathematically concise and elegant manner that it was a real pleasure to read.

I am, however, a little unsure about the value of the paper to the average practising data analyst (not necessarily a trained statistician) who may be faced with problems similar to those described. One may hope that the examples presented would enable such a person to recognize that his particular problem can be solved by the use of the technique but in this paper there is little to show him exactly how it can be done. I may have been misled by the title of course to expect something a little more practical, which dealt, for example, with methods of speeding convergence and of calculating the increased variance due to the missing information. With regard to this hidden variance it is my view that in recent years papers dealing with missing-data techniques have paid too little attention to it. As a result, I have come across people who, when using an analysis of variance package, were content to analyse data, 30 per cent of which was missing, under the assumption that it made no difference to the calculated test statistics. The package dealt with the estimation problem in an extremely efficient manner but completely ignored the increased variance. I should therefore like the authors to comment on the ease with which the hidden variance can be calculated using the EM algorithm.

An item in the paper which I found to be particularly interesting was the concept of using an informative prior to provide additional information and I would like to know if the speakers have ever applied this in a practical problem. In connection with this I was interested to hear Professor Beale speak of using imputed values since I too have been giving some thought to the effect of using imputed values, in my case for census and survey data. I wonder then if the authors have ever had any thoughts on the use of "hot-deck" and "cold-deck" procedures since these can be regarded as using prior information, and in some cases an approach similar to the EM algorithm.

Mr B. TORSNEY (University of Glasgow): The authors have illustrated the application of their EM algorithm in a wide range of statistical problems. The following may be one further example in an optimal design problem.

Problem. Choose J weights p_i subject to $p_i \geq 0$, $\sum p_i = 1$, the summation being from 1 to J ,

$$\phi(\mathbf{p}) = -\text{tr}(\mathbf{M}^{-t}), \quad \mathbf{M} = \sum_{i=1}^J p_i \mathbf{v}_i \mathbf{v}_i^T, \quad t > 0. \quad (1)$$

The $\mathbf{v}_i$'s are the known $k \times 1$ vectors forming the design space. $\phi(\mathbf{p})$ is (a) increasing

$$(\partial\phi/\partial p_i = t\mathbf{v}_i^T \mathbf{M}^{-(t+1)}, \mathbf{v}_i > 0),$$

(b) homogeneous of degree $-t$, (c) concave.

The following algorithm, suggested by a result in Fellman (1974), has been formulated by Silvey *et al.* (1976):

$$p_i(r) \propto p_i(r-1) \{ \partial\phi/\partial p_i(r-1) \}^\delta, \quad \delta > 0.$$

The $p_i(r)$ are normalized and so stay in the feasible region.

If any $\phi(\mathbf{p})$ possesses (a) this increases $\phi(\mathbf{p})$ when δ is small, while for any $\delta > 0$ and small $\varepsilon > 0$, $\phi(\mathbf{q}) \geq \phi(\mathbf{p}(r-1))$ where $\mathbf{q} = (1-\varepsilon)\mathbf{p}(r-1) + \varepsilon\mathbf{p}(r)$.

However, various considerations suggest that $\delta = 1/(t+1)$ is a natural power to use in (1); in simple cases this attains the optimum in one step.

It can be shown that if $\phi(\mathbf{p})$ possesses (a) and (b) then $\delta = 1/(t+1)$ achieves monotonicity if

$$\psi_\mu(\lambda) \geq \psi_\mu(\mu), \quad (2)$$

where

$$\lambda = \mathbf{p}(r-1), \quad \mu = \mathbf{p}(r), \quad \psi_\mu(\lambda) = \sum_{i=1}^J f_{i\mu}(\lambda), \quad f_{i\mu}(\lambda) = \lambda_i (\partial\phi/\partial \lambda_i)^{1/(t+1)} (\partial\phi/\partial \mu_i)^{t/(t+1)}.$$

This in turn is true by Baum *et al.* (1970) if $\phi_\mu(\lambda|\mu) \geq \phi_\mu(\mu|\mu)$ where $\phi_\mu(\lambda|\mu) = \sum f_{i\mu}(\lambda) \log f_{i\mu}(\lambda)$, summing between 1 and J .

Results so far established are that for any pair λ, μ (assuming $\partial\phi/\partial \lambda_i, \partial\phi/\partial \mu_i$ exist)

(i) (2) is true when $\phi(\mathbf{p}) = -\text{tr}(\mathbf{M}^{-1})$.

(ii) Both $\psi_\mu(\lambda)$ and $Q_\mu(\lambda|\mu)$ have stationary values at $\lambda = \mu$ if $\phi(\mathbf{p})$ possesses (a) and (b).

Hence there is the possibility that $\delta = 1/(t+1)$ achieves monotonicity for several such functions. However, counter-examples exist. Other properties of $\phi(\mathbf{p})$ can be relevant to a natural choice of δ . For example, $\phi(\mathbf{p}) = \det(\mathbf{M})$, though not concave, is a homogeneous polynomial of degree k with positive coefficients. Thus $-\{\det(\mathbf{M})\}^{-1}$ possesses (a) and (b) with $t = k$, but $\delta = 1$ emerges as a natural power since this attains the optimum in one step if $J = k$; it achieves monotonicity by Baum and Eagon (1967). This power, however, is a special case of $\delta = 1/(t+1)$ above since $\{\text{tr}(\mathbf{M}^{-t})/k\}^{1/t} \rightarrow \{\det(\mathbf{M})\}^{-1/k}$ as $t \rightarrow 0$ and $\partial \det(\mathbf{M})/\partial p_i = (\mathbf{v}_i^T \mathbf{M}^{-1} \mathbf{v}_i) \det(\mathbf{M})$. The resultant algorithm can in fact be shown directly to be an EM algorithm. Can this be done for $\phi(\mathbf{p})$ in (1)?

On the question of naming the authors' technique; north of the border we would be content to describe it as a wee GEM!

Dr D. M. TITTERINGTON (University of Glasgow) and Dr B. J. T. MORGAN (University of Kent): Speed of convergence is an important factor in judging algorithms. We have recently considered the hypothesis of quasi-independence in an $m \times m$ contingency table with a missing main diagonal. If $\{a_i\}$ and $\{p_j\}$ represent the row and column probabilities, it turns out that there is choice about which version of the EM algorithm to use for estimating parameters, one depending in the E-step on both sets of parameters and one each on the $\{a_i\}$ and $\{p_j\}$. Alternate use of the last two corresponds to Goodman's (1968) version of iterative scaling. If, however, only the last is used, thereby reducing the number of parameters involved in the E-step, it appears empirically that convergence is usually faster; see Morgan and Titterington (1977). Further acceleration can often be obtained using an iteration of the form

$$p_j^{(r)} \propto p_j^{(r-1)} \left(\frac{\partial \phi}{\partial p_j^{(r-1)}} \right)^\delta, \quad j = 1, \dots, m, \quad r \geq 1, \quad (1)$$

where $\delta \geq 1$ and ϕ is the log-likelihood for the *observed* data, evaluated at $\{p_j^{(r-1)}\}$. Often $\delta > 1$ produces faster convergence than the basic EM algorithm, given by $\delta = 1$; see also Mr Torsney's remarks and Silvey *et al.* (1976), where an iteration like (1) is used in computing D -optimal designs.

In the examples we looked at, easily the quickest method was that of Brown (1974), although it may not always be monotonic for ϕ , in which the missing cells are treated one by one. This behaviour seems similar to the superiority, in certain examples, of Fedorov's (1972) algorithm over the appropriate EM method (Silvey *et al.*, 1976) for computing D -optimal designs. In each iteration of the former a single design weight is changed optimally, instead of changing all weights at once in a non-optimal way. We wonder if the idea of reducing each EM iteration to a sequence of stages, each of which is an exact maximization in some sense, may accelerate convergence in other applications.

Mr GORDON D. MURRAY (University of Glasgow): I have been using the EM algorithm a great deal recently to estimate the parameters of multivariate normal distributions using incomplete samples (Section 4.1.3), and I have found that a considerable practical problem is the existence of multiple stationary values. The following example illustrates the kind of problems which can arise.

Suppose that we have the following data, representing 12 observations from a bivariate normal population with zero means, correlation coefficient ρ and variances σ_1^2, σ_2^2 .

Variable 1	1	1	-1	-1	2	2	-2	-2	*	*	*	*
Variable 2	1	-1	1	-1	*	*	*	*	2	2	-2	-2

The asterisks represent values which were not observed. For these data the likelihood has a saddle point at $\rho = 0$, $\sigma_1^2 = \sigma_2^2 = \frac{5}{2}$, and two maxima at $\rho = \pm \frac{1}{2}$, $\sigma_1^2 = \sigma_2^2 = \frac{8}{3}$.

The EM algorithm will converge to the saddle point from a starting point with $\rho = 0$, but, as the authors point out, this will not be a problem in practice, because a random perturbation will cause the algorithm to diverge from the saddle point. The example is also rather artificial in that the two local maxima have equal likelihoods. In general there will be a unique global maximum.

I have run the EM algorithm on many simulated data sets, using two different starting points: (1) taking zero means and identity covariance matrix and (2) taking the maximum-likelihood estimates based only on the complete observations. The results obviously depend on the pattern of missing data, but it is not unusual for about 5 per cent of the cases to converge to different local maxima. This technique of course only gives a lower bound on the occurrence of multiple stationary points.

This may seem rather removed from real life, but the study was in fact motivated when this problem arose while I was working on a set of real eight-dimensional data. I obtained two sets of estimated parameters which were essentially the same, except for the estimates of one of the variances, which differed by a factor of 30!

This is naturally not a criticism of the algorithm itself, but it should be a warning against its indiscriminate application.

Dr. D. A. PREECE (University of Kent at Canterbury): Dr Nelder has mentioned the advantage of introducing the multiplier N/ν into the Healy–Westmacott procedure. But I know of no enumeration of the exceptional circumstances in which the configuration of missing values and the form of the analysis are such that the procedure with N/ν fails to converge. G. N. Wilkinson has told me of an example of such non-convergence, and I have seen a rather degenerate example derived by someone else. But I do not know of any theoretical formulation showing exactly what such examples are possible. If tonight's authors—or anybody else—could throw light on this, I for one should be grateful.

I should be embarrassed if my name came to be associated with the algorithm incorporating the N/ν . It seems that different people hit on this *improved Healy–Westmacott procedure*; I was not one of them. I think the earliest account of it is in a book by Pearce (1965, pp. 111–112), and I understand that Professor Pearce thought of it some years before the book appeared. When I wrote my 1971 paper I was, I regret, unaware that Professor Pearce had obtained the improved procedure.

The multiplier N/ν for a two-way analysis also figures in the FUNOR–FUNOM procedure of Tukey (1962, pp. 24–25).

Dr KEITH ORD (University of Warwick): The authors are to be congratulated upon their bravery in discussing estimation methods for factor analysis, a topic which still causes some statisticians to blow a fuse. One of the difficulties in such situations is that the standard parametrization allows arbitrary orthogonal transformations of the matrix of loadings, β , which do not affect the value of the likelihood. To avoid this, previous researchers have found it necessary to impose constraints such as

$$\beta^T(\tau^2)^{-1}\beta = \mathbf{J},$$

where $\mathbf{J}$ is an arbitrary diagonal matrix. I would be interested to learn how the authors method is able to avoid such constraints and the computational problems that arise in their wake.

Mr M. J. R. HEALY (Medical Research Council): On the question of speed of convergence, the EM technique as applied to designed experiments is numerically equivalent to the Jacobi method of solving simultaneous linear equations. This is known to converge more slowly than the Gauss–Seidel method which is in its turn equivalent to the traditional missing-value technique of adjusting the replaced values one at a time. There is a very large literature on speeding the convergence of the Gauss–Seidel method by “stretching” corrections, under the name of over-relaxation; it would be interesting to see whether this could be applied to the Jacobi method.

The suggested method for factor analysis can be related to that published by Rao (1955). In this method, canonical correlations between the observed variables and the “missing” factor scores are calculated at each stage of an iteration. Conventional canonical analysis can be calculated by an iterative back-and-forth regression technique, and the authors' method can be regarded as steps in an inner iterative loop. In such cases it often makes excellent sense not to drive the inner iteration to completion. My own very limited experience of Rao's method (and equally those of Howe, 1955, and Bargmann, 1957) is that convergence is impossibly slow—the iterative corrections are small, but they do not seem to get any smaller as the iteration progresses. Could this be due to the arbitrary rotation that is involved?

The following contributions were received in writing after the meeting.

Mr LEONARD E. BAUM (Institute for Defense Analysis, NJ, USA): In the penultimate paragraph of Section 3, the authors write:

"Lemma 1 and its consequence Theorem 1 were presented by Baum *et al.* (1970) in an unusual special case (see Section 4.3 below), but apparently without recognition of the broad generality of their argument."

In Baum *et al.* (1970) we have: let

$$P(\lambda) = \int_{\mathcal{X}} p(x, \lambda) d\mu(x) \quad \text{and} \quad Q(\lambda, \lambda') = \int_{\mathcal{X}} p(x, \lambda) \log p(x, \lambda') d\mu(x).$$

Theorem 2.1. If $Q(\lambda, \bar{\lambda}) \geq Q(\lambda, \lambda)$ then $P(\bar{\lambda}) \geq P(\lambda)$. The inequality is strict unless

$$p(x, \lambda) = p(x, \bar{\lambda}) \quad \text{a.e. } [\mu].$$

With the change of variables and notation $\lambda \rightarrow \Phi$, $\bar{\lambda} \rightarrow \Phi'$, $\log P(\lambda) \rightarrow L(\Phi)$, $p(x, \lambda) \rightarrow f(x | \Phi)$, $\log p(x, \lambda') \rightarrow \log f(x | \Phi')$, $d\mu(x) = d(x)$ for $x \in \mathcal{X}(y)$, = 0 otherwise, our $Q(\Phi, \Phi')$ equals the $g(y, \Phi) Q(\Phi' | \Phi)$ of this paper and hence our Theorem 2.1 and its use with a transformation τ (= the EM algorithm) in our Theorem 3.1 contains the present paper's Lemma 1 and Theorem 1.

In the numerous examples of Section 4 of this paper the unseen sample space $\mathcal{X}$ is a sample space of independent variables so $g(y | \Phi)$ is essentially of the form

$$g(y | \Phi) = \prod_{i=1}^T \int_{\mathcal{X}(y_i)} f(x_i | \Phi) dx_i.$$

In our papers we considered the case $\mathcal{X}$ a Markov sample space which contains the case $\mathcal{X}$ independent as a special case. In the Markov case

$$g(y | \Phi) = \int_{\mathcal{X}(y_1)} \dots \int_{\mathcal{X}(y_T)} \prod_{i=0}^{T-1} f(x_i, x_{i+1} | \Phi) dx_1 \dots dx_T$$

is not so simple as in the independent case so an additional inductive algorithm is required for effective computation of the E step. See the second, third and fourth references listed in this paper.

Professor W. H. CARTER (Virginia Commonwealth University): Professors Dempster, Laird and Dr Rubin are to be commended on the presentation and thorough treatment of an interesting problem.

As a result of this paper, renewed interest in this algorithm will be generated and numerous programs will be written. The difficulties associated with obtaining the properties of maximum-likelihood estimators when the estimator cannot be written in closed form are well known. I should like to point out that Hartley (1958) indicated a numerical method of estimation based on the calculus of finite differences which could be used to obtain variance and covariance estimates of the maximum-likelihood estimates obtained by the EM algorithm. Basically, the procedure involves estimating the second derivatives of the log-likelihood function from the iterations made to determine the root(s) of the likelihood equation(s). Hartley illustrates the method for a single parameter and a multiparameter distribution.

Clearly, the advantage of such a procedure is that it can be incorporated in the computer program written to obtain the maximum-likelihood estimates of the parameters so that the final program produces, simultaneously, the estimates and estimates of their variances and covariances.

Professor B. EFRON (Stanford University): This is an excellent paper, difficult for me to criticize on almost any grounds, which is fine and good for the authors, but hard on potential critics. I will settle for an historical quibble. Let $DL^*(\Phi)$ indicate the Fisher score function based on the complete data set $\mathbf{x}$, that is the derivative of the log density with respect to the parameter, and let $DL^y(\Phi)$ be the score function based on some statistic $y(\mathbf{x})$. In his 1925 paper Fisher showed that

$$DL^y(\Phi) = E_{\Phi}(DL^*(\Phi) | y). \quad (1)$$

(See p. 717, where the result is used, though in typical Fisherian fashion not explicitly mentioned in calculating the loss of information suffered in using an insufficient statistic.)

For the exponential family (2.1), $DL^*(\Phi) = t - E_{\Phi}(t)$. In this case equation (1) becomes

$$DL^y(\Phi) = E_{\Phi}(t | 0) - E_{\Phi}(t) \quad (2)$$

which is the "striking representation" (2.13).

Professor STEPHEN E. FIENBERG (University of Minnesota): It is a great pleasure for me to have an opportunity to discuss this interesting and important paper. It not only presents a general approach for maximum-likelihood estimation in incomplete-data problems, but it also suggests a

variety of ways to adapt the approach to complete-data problems as well. I regret being unable to hear its presentation, although I did hear informal lectures on the topic by two of the authors about one year ago. One of the reasons I was delighted to see these results is that they touch on so many seemingly unrelated problems I have worked on in the past, and on several that sit in various stages of completion on my office desk.

In 1971, Haberman noted that the research work on categorical data problems of two of my graduate students at the University of Chicago could be viewed in a more general context as problems involving frequency tables based on incomplete observation. One of these students was obviously working on a missing-data problem but the other student's work had been developed as a complete-data problem. By using a representation similar to that used by Dempster, Laird and Rubin in the genetics example of Section 1, Haberman showed how seemingly complete data can often be represented as incomplete data. The present paper shows that this approach is applicable in far more general situations. It should come as no surprise that at least one of the iterative methods proposed by Haberman can be viewed as using the EM algorithm. Haberman (1974), in extending his earlier work, noted two problems in the case of frequency data which have a direct bearing on Theorem 2 of the present paper: (a) even for cases where the likelihood for the complete-data problem is concave, the likelihood for the incomplete problem need not be concave, and (b) the likelihood equations may not have a solution inside the boundary of the parameter space. In the first case multiple solutions of the likelihood equations can exist, not simply a ridge of solutions corresponding to a lack of identification of parameters, and in the second case the solutions of the likelihood equations occur on the boundary of the parameter space, which is usually out at infinity if we consider the problem in terms of the natural parameters ϕ .

The problem of a solution on the boundary comes up not only in categorical data problems but also in factor analysis as considered in Section 4.7. There the boundary problems are referred to as *improper solutions* or Heywood cases, and correspond to zero values for one or more of the residual variances τ^2 . In practice, when one is doing factor analysis, it is important to use an algorithm that detects these improper solutions after only a few iterations, so that a revised iteration can be initiated using the zero estimates initially. The Jöreskog algorithm described in Lawley and Maxwell (1971) is specifically designed to handle such problems, while it appears that the version of the EM algorithm outlined in Section 4.7 may be stopped long before a Heywood case can be recognized. Have the authors explored this facet of the problem? Factor analysis in situations with small sample sizes also presents examples where multiple solutions to the likelihood equations exist.

While problems do occur in the use of the EM algorithm, the general formulation of Dempster, Laird and Rubin is remarkable in that it leads to an incredibly simple proof of the convergence properties. When I worked with a special case of the algorithm (see Chen and Fienberg, 1976), my co-author and I were unable to deal properly with the convergence properties. All we were able to show was that our procedure converged if at some stage our estimate was sufficiently close to the true solution and if the sample size was large. By generalizing the problem the present authors have made the problems we encountered disappear!

The authors discuss aspects of the rate of convergence of the EM algorithm at the end of Section 3, but they do not discuss its computational efficiency relative to other algorithms in specific cases when iteration is in fact necessary. In many applications the procedure is computationally superior to all competitors. In others, however, the EM algorithm is distinctly worse in performance when compared with one or more alternative algorithms. For example, when the version of the EM algorithm proposed by Brown (1974) is used for estimation in the model of quasi-independence in square $I \times I$ contingency tables with missing diagonals, it is distinctly superior to the standard iterative proportional fitting algorithm described in Bishop *et al.* (1975). Yet, when the number of missing cells in the $I \times I$ is large, the iterative proportional fitting procedure is more efficient than the EM algorithm. The beauty of the EM algorithm is its simplicity and generality; thus, we should not be surprised at its inefficiency in particular problems. (A similar comment is appropriate for the use of iterative proportional fitting as an all-purpose algorithm for contingency table problems.)

I was especially pleased to see Section 4.5 dealing with hyperparameter estimation since it appears to be of use in a problem I am currently exploring, that of characterizing linear forecasts and their extensions resulting from the use of multi-stage prior distributions, and approximations to the latter using estimates of the hyperparameters. Now that the authors have taught us how to recognize such nonstandard applications of their approach, I am sure that their work will lead to many new applications in other active areas of statistical research.

Professor IRWIN GUTTMAN (University College London and University of Toronto): In practice, the idea of the EM algorithm has been used in various places other than those indicated by the authors, through the use of the predictive distribution. The predictive distribution of y , given the data x , is defined as

$$p(y|x) = \int f(y|\theta) p(\theta|x) d\theta \quad (1)$$

where $p(\theta|x)$ is the posterior of the parameter θ that controls the distribution of x and the future observations y , namely $f(\cdot|\theta)$. If a data-gathering process D is used to obtain x , while the data-gathering process D' is to be used to generate y , we denote the predictive distribution as

$$p(y, D'|x, D) = \int f(y|\theta; D') p(\theta|x; D) d\theta. \quad (1a)$$

Suppose germane to this problem, the utility function of the act t taken with respect to θ is $u(t, \theta)$. Define the average posterior utility as

$$U(x, y; D, D') = \int u(t(x; y); \theta) p(\theta|x, D; y, D') d\theta \quad (2)$$

where we have acted in (2) as if all the observations x and y are now at hand. The *expected part of the predictive algorithm* says to find

$$E(U|x, D) = \int U p(y, D'|x, D) dy = g(x, D; D') \quad (3)$$

and the *maximization part of the predictive algorithm* says to choose D' so as to maximize g .

The applications have been many and varied—as two examples amongst a host of others, see Draper and Guttman (1968) for its use in an allocation problem in sample survey, and Guttman (1971) for an application involving optimum regression designs. Indeed, the above is just another way of looking at some aspects of what has been called *preposterior* analysis by Raiffa and Schlaifer (1961). The emphasis here is not on parameters, but on predictions (estimation of observations) that have optimal properties. Indeed, the above framework allows for missing data, truncation, censoring, etc. through the flexible use of the definition of D . Indeed in a forthcoming paper by Norman Draper and myself, it has been used in a particular missing-value problem.

An important point to note here is that the form of g given in (3) may or may not be such that sufficient statistics are estimated in the expected part of the predictive algorithm—in principle, the above goes through without the need for sufficiency.

Dr S. J. HABERMAN (University of Chicago): The authors must be congratulated on their wide-ranging discussion of the EM algorithm. Although I share their admiration for the versatility and simplicity of the procedure, I have ambivalent feelings toward its use. The Newton–Raphson and scoring algorithms are competing numerical procedures with both advantages and disadvantages relative to the EM algorithm.

In favour of the EM algorithm are simplicity in implementation and impressive numerical stability. However, the Newton–Raphson and scoring algorithms are not especially difficult to implement, and they do provide estimates of asymptotic variances. In addition, convergence of the EM algorithm is often painfully slow, based on my experience with latent-structure models. In contrast, the Newton–Raphson and scoring algorithms generally result in rapid convergence.

The Newton–Raphson and scoring algorithms can be described in terms of the first two conditional and unconditional moments of t given ϕ , just as the EM algorithm can be described in terms of the corresponding first cumulants. If $\phi^{(p)}$, $p \geq 0$, is the sequence of approximations generated by the algorithm, then by the authors' equation (2.16), the Newton–Raphson algorithm may be defined by the equation

$$\phi^{(p+1)} = \phi^{(p)} + \{V(t|\phi^{(p)}) - V(t|y, \phi^{(p)})\}^{-1}\{E(t|y, \phi^{(p)}) - E(t|\phi^{(p)})\},$$

and the scoring algorithm may be defined by the equation

$$\begin{aligned} \phi^{(p+1)} &= \phi^{(p)} + [V(t|\phi^{(p)}) - E\{V(t|y, \phi^{(p)})|\phi^{(p)}\}]^{-1}\{E(t|y, \phi^{(p)}) - E(t|\phi^{(p)})\} \\ &= \phi^{(p)} + [V\{E(t|y, \phi^{(p)})|\phi^{(p)}\}]^{-1}\{E(t|y, \phi^{(p)}) - E(t|\phi^{(p)})\}. \end{aligned}$$

Here $E\{V(t|y, \phi^{(p)})|\phi^{(p)}\}$ denotes the expected value of $V(t|y, \phi^{(p)})$ when y has sampling density $g(y|\phi)$. A similar convention applies to $V\{E(t|y, \phi^{(p)})|\phi^{(p)}\}$.

The asymptotic covariance matrix of ϕ^* may be estimated by

$$\{V(t|\phi^*) - V(t|y, \phi^*)\}^{-1}$$

if the Newton–Raphson algorithm is used or by $[E\{V(t|\phi^*)|\phi^*\}]^{-1}$ if scoring is used.

Thus many of the attractive relationships between algorithms and moments of the EM algorithm are retained by the older Newton–Raphson and scoring algorithms.

I assume that the major criterion in a decision to use the EM algorithm should be the extent to which estimates of asymptotic variances and covariances are needed. If these estimates are clearly needed, then I suspect the EM algorithm is relatively less attractive than if such estimates are only of marginal interest. I am curious how the authors view the relative merits of these procedures.

Professor H. O. HARTLEY (Texas A & M University): I feel like the old minstrel who has been singing his song for 18 years and now finds, with considerable satisfaction, that his folklore is the theme of an overpowering symphony. However, I wish the authors would have read my “score” more carefully (and I use “score” also in the Fisherian sense). The “score” that I was singing then (Hartley, 1958, pp. 181–182) is not confined to the multinomial distribution but covers *any* grouped discrete distribution although a binomial example preceded the completely general formulae as an illustration. Incidentally, the case of truncated discrete distributions for which I developed (pp. 178–179) an algorithm analogous to the EM algorithm does not fall within the framework of the author’s incomplete data specification (their equation (1.1)). Since it is an essential feature of any “algorithm” that its formulae can be numerically implemented, I confined my 1958 paper to discrete distributions (when the E operator is a simple sum) but in the 1971 paper (with R. R. Hocking) we extended the E operator to continuous distributions with the help of formulae for numerical integration. The authors’ formulation of an incomplete-data likelihood is certainly more comprehensive. On the other hand, they do not discuss the feasibility of the numerical implementation of the algorithms. (See, for example, the general E-step which involves the computation of an r -parametric function depending on ϕ' using a conditional likelihood on ϕ, y .)

Their Theorems 2–4 specify convergence conditions for the EM algorithm which are more restrictive than the one given by us in our 1971 paper (pp. 806–808). Specifically the authors assume that the eigenvalues of their $D^{20} Q(\phi^{(p+1)}|\phi^{(p)})$ are bounded away from zero. No such assumption is made in our proof and in fact usually such an assumption is certainly not satisfied identically in the parameter space except for the exponential family quoted by them. Their references to “ridges” are not clear. For if we define “ridges” by the existence of a function $h(\phi_1, \dots, \phi_k)$ of (say) the first $k \geq 2$ elements of ϕ such that

$$f(x|\phi) = f(x|h(\phi_1, \dots, \phi_k), \phi_{k+1}, \dots, \phi_r)$$

then the above matrix D^{20} has a rank $\leq r - k + 1 < r$ and its eigenvalues are clearly not bounded away from zero. This is a consequence of the authors’ equation (3.13) and of the interchangeability of the operators $E|y, \phi^{(p)}$ and D^{10} , D^{20} operating on $\log f(x|\phi)$. Such “ridges” would, of course, also violate the main assumption we have made namely that $\lambda = \log g(y|\phi)$ cannot have two separate stationary points in the ϕ -space with identical values of λ . However, this latter condition is only violated with probability zero if ridges of the above type are excluded.

Finally, I would like to draw the authors’ attention to our method of variance estimation (pp. 185–188 in Hartley, 1958; pp. 796–798 in Hartley and Hocking, 1971) which utilizes the iterates in the EM algorithm directly for the estimation of the elements of the variance matrix and this potential of the EM algorithm is important in justifying its computational efficiency compared with competitive ML estimation procedures.

Professor S. C. PEARCE (University of Kent): I share the general pleasure at the width of application of this paper but I join with those who fear possible slowness of convergence of the EM algorithm. As Dr Nelder has pointed out, one case where the situation is well explored is the fitting of missing values in a designed experiment. As I understand him, the algorithm gives the well-known method of Healy and Westmacott (1956), which always converges, though it can be very slow. The accelerated method in which the residual is multiplied by n/v has been known for a long time by many people, as Dr Nelder and Dr Preece remark, and pathological cases are known in which it will lead to divergence. However, at East Malling we used it as a regular procedure in all computer programs written from about 1961 onwards and I cannot recall any instance of its

having diverged in practical use, at least not before I left early in 1975. I do not know what has happened since. Anyhow, there is another accelerated method that does ensure convergence in a single cycle if only one plot is missing. I refer to the use of the multiplier, $1/e$, where e is the error sum of squares from the analysis of variance of p , a pseudo-variate having one for the plot in question and zero for all others (Pearce and Jeffers, 1971; Rubin, 1972). It shows that the EM algorithm, though of general application, is not optimal.

Professor S. R. SEARLE (Cornell University): My comments are confined largely to Section 4.4, dealing with variance components. Several phrases there are strange to the variance components literature: (i) "making the $x_1, x_2, \dots, x_k$ also normally distributed ... converts the model ... to a random effects model": random models are not necessarily normal; (ii) "compute the mean ... of the conditional normal distribution of the x_i given y ": why will it be anything other than null, in view of 4.4.4 and 4.4.5? y of 4.4.1 needs a mean μ ; (iii) "where ($x_2, x_3, \dots, x_{k+1}$) are regarded as missing": how can they be when their variances are being estimated?—unobservable, yes, but surely not missing; (iv) "except that σ_1 is fixed at ∞ ": fixed effects usually have zero, not infinite, variance.

The authors' application of EM to estimating variance components displays no advantages for EM over methods described by Hemmerle and Hartley (1973) and Hemmerle and Lorens (1976), which are directly suited to variance components models and take account of their special properties—whereas EM does not. In particular, EM pays no attention to computational difficulties brought about by the r_i being very large in many of the data sets from which variance components estimates are required. Hemmerle and Lorens (1976), for example, show how to discount this effect by a factor of 4.

General properties of EM are described in Section 3. It is a pity that their usage and importance to special cases were not indicated for the examples of Section 4.

Dr R. SUNDBERG (Royal Institute of Technology, Stockholm): I want to congratulate the authors on an important and brilliant paper. The great value of their generalizations and unifications is obvious, and the paper will certainly stimulate more frequent use of the EM algorithm. But because of this, a warning against uncritical faith in the method may be appropriate.

After some initial steps the deviation from a local maximum point decreases at each step by a factor which approximately equals the largest eigenvalue, $\lambda_{\max}$ say, of (3.15) (or (3.27)), expressing the maximal relative loss of information due to incompleteness. In applications $\lambda_{\max}$ is often close to 1, and in my own experience, from applications to mixtures and to convolutions, $\lambda_{\max}$ has sometimes turned out to be so close to 1 (> 0.98 , say) that I have judged the method to be impracticable. For instance, with data from a five-parametric mixture of two normals it can be seen from Fig. 3 in Sundberg (1976) that the EM method will require a very large number of steps when $(\sigma_1 + \sigma_2)/|\mu_1 - \mu_2| > 1$. When this occurs the formulation of the stopping rule is crucial and numerical extrapolation is difficult. It does not mean that the ML estimation principle has broken down in these cases, only that very large samples have been required for precise estimates.

The Newton–Raphson method and the scoring method do not have this disadvantage for large values of $\lambda_{\max}$, but instead they entail matrix inversion problems. However, the inverse of the information matrix is anyhow desired when the final estimates have been attained. Therefore I want to advocate the use of these two methods when $\lambda_{\max}$ seems to be close to 1.

Dr E. A. THOMPSON (University of Cambridge): I would like to mention a couple of cases where the EM algorithm arises in genetics. One is the classical "gene-counting" method of estimating allele frequencies from phenotype frequencies. The number of genes of each allelic type are estimated using the phenotype numbers and assumed allele frequencies, and new allele frequencies are then calculated from these numbers. This is essentially a case of the multinomial example discussed in the paper.

A more interesting example is given by the problem of estimating evolutionary trees from population allele frequencies. I refer to the modification of the model of Edwards (1970) discussed by Thompson (1975). There we have a bifurcating tree model of population evolution, and, as the population allele frequencies change under the action of random genetic drift, the populations perform Brownian motion in a Euclidean space, in which the co-ordinates are certain functions of these allele frequencies. One part of the problem is to estimate the times of split and the position of the initial root, given the final population positions and assuming a given tree form.

On p. 68 of Thompson (1975) an iterative solution is proposed. Although at each cycle the maximization with regard to the root position is carried out explicitly in terms of the data and current estimates of the splitting times, the method proposed for estimating these times is precisely an EM algorithm. If the positions in the gene-frequency space of all populations at the instants of splitting of any single population are regarded as the missing data, we have, for a *given* root position, a particularly simple regular exponential family. The natural parameters are the inverses of the time intervals between splits, the sufficient statistics are the "total squared distances travelled in that time interval by all populations then existent" (p. 64), and their unconditioned expectations are simple multiples of the time intervals to be estimated.

I wish I had known of the EM algorithm in 1972; it would have greatly aided my discussions of uniqueness and convergence problems, at least in those cases where a root in the interior of the parameter space exists.

Mr ROBIN THOMPSON (ARC Unit of Statistics, Edinburgh): Whilst schemes such as (4.4.8) are very appealing for variance component estimation their convergence can be painfully slow and I have found that schemes using second derivatives are more satisfactory. The theory of Section 3 is useful in explaining why the convergence can be slow. Consider a random effects model for a one-way classification with n observations in each group and let γ denote the ratio of the between group variance component to the within group variance. The largest root of $\mathbf{DM}(\phi^*)$ is approximately $\max [1 - \{\gamma/(n\gamma + 1)\}^2, 1/n]$. This tends to 1 as $n\gamma$ tends to zero. On the other hand, if $n\gamma$ is large the largest root is of the order $1/n$ and estimation using (4.4.8) should converge rapidly.

Professor B. TURNBULL (Cornell University): The authors have provided a very elegant and comprehensive treatment of maximum-likelihood estimation with incomplete data. I regret that I arrived in London one week too late to hear the presentation and discussion.

Related to the problems of estimation are those of goodness-of-fit with incomplete data. They appear to be especially important in reliability and recidivism studies. For grouped and randomly censored data, Lionel Weiss and myself (1976) have proposed a likelihood ratio test. In the numerator of the ratio appears the maximum likelihood under the postulated family of parametric models, and the denominator contains the likelihood based on the empirical distribution function. Both likelihoods are calculated using the EM algorithm or, equivalently, "self-consistency" (Turnbull, 1976). The resulting statistic is shown to have an asymptotic chi-squared distribution with the appropriate non-centrality parameter under contiguous alternatives. The test is applied in a study of marijuana usage in California high schools where the data are both grouped and doubly censored.

Perhaps the lesser known papers of Batschelet (1960) and Geppert (1961) should be added to a list of references concerning maximum-likelihood estimation in incomplete contingency tables.

It should be noted that in many problems it is hard to justify the assumption that the censoring mechanism is independent of the data (observed or unobserved), e.g. losses to follow-up in medical trials. In such cases of "prognostic censoring", it seems that little can be done (Tsiatis, 1975; Peterson, 1975). Presumably two "extreme" analyses, one optimistic and one pessimistic, could be performed. Similar dependent censoring can occur in cross-over studies when a subject, who is faring poorly on the placebo, is switched onto the treatment prematurely.

The authors replied in writing, as follows:

We thank the many discussants for their stimulating and constructive remarks. Our response is mainly organized around themes developed by each of several discussants.

Many discussants express interest in speeding the rate of convergence of the EM algorithm. Dr Nelder, Dr Preece, Dr Pearce and Mr Healy point out that the rate of convergence of the EM can often be improved in the special case of missing values in ANOVA by "stretching" procedures, although apparently at the cost of sacrificing sure convergence (Hemmerle, 1974, 1976). We suggested in Section 3 that Aitken's acceleration routine may be useful, and Professor Smith suggests a method of improving on Aitken's routine for a single parameter. How to implement such methods routinely in multi-parameter problems is not clear.

Dr Little gives an interesting example which demonstrates that the choice of "complete" data can influence the rate of convergence, the reason being that reducing the fraction of missing information speeds convergence. Specifically,

$$\mathbf{DM}(\phi^*) = \mathbf{D}^2 H(\phi^* | \phi^*) \{ \mathbf{D}^2 L(\phi^*) + \mathbf{D}^2 H(\phi^* | \phi^*) \}^{-1},$$

where $\mathbf{D}^2 L(\phi^*)$ is fixed by the incomplete-data specification and the observed value $\mathbf{y}$, but $\mathbf{D}^2 H(\phi^* | \phi^*)$ is influenced by the method of completing the data. A judicious choice of complete data will (a) reduce the maximum eigenvalue of $\mathbf{DM}(\phi^*)$, and (b) allow easy computation of the E- and M-steps. Unfortunately (a) and (b) may work at cross-purposes, as in the case of a truncated sample from a normal population, where treating the number of truncated observations as missing slows convergence but greatly eases the M-step.

We are indebted to Mr R. Thompson for drawing attention to a vexing situation which comes up often in variance components estimation, namely, when a maximum-likelihood estimate of some variance component is zero, the rate of convergence of EM also goes to zero. A similar difficulty arises in estimating uniquenesses in factor analysis. These uniquenesses are also variance components, and when their estimates go to zero, the situation is known as the Heywood case (cf., Professor Fienberg's discussion). A heuristic explanation of the vanishing rate of convergence in these examples is that as a variance goes to zero, the information about the variance also goes to zero, and this vanishing information implies a vanishing rate of convergence.

As suggested by Professor Haberman, Dr Little and Dr Sundberg, it is important to remember that Newton-Raphson or Fisher-scoring algorithms can be used in place of EM. The Newton-Raphson algorithm is clearly superior from the point of view of rate of convergence near a maximum, since it converges quadratically. However, it does not have the property of always increasing the likelihood, and can in some instances move towards a local minimum. Consequently, the choice of starting value may be more important under Newton-Raphson. In addition, the repeated evaluation and/or storage of the second derivative matrix can be infeasible in many problems. For complete-data problems the scoring algorithm will be equivalent to Newton-Raphson if the second derivative of the log-likelihood does not depend on the data (as with exponential family models). In these cases, the scoring algorithm also has quadratic convergence. However, scoring algorithms often fail to have quadratic convergence in incomplete-data problems since the second derivative often does depend upon the data even for exponential family complete-data models.

One advantage of Newton-Raphson or Fisher-scoring is that an estimate of the asymptotic covariance matrix is a by-product of the computation of ϕ^* . Professors Carter and Hartley speak to a question raised by Mr Orchard in remarking that Hartley and Hocking (1971) noted the possibility of obtaining an estimate of the asymptotic covariance matrix from successive iterates of the EM algorithm. As Professor Smith notes, an estimated asymptotic variance is readily obtained in the single parameter case as

$$\hat{\sigma}_{\phi^*}^2 / (1 - \lambda),$$

where $\hat{\sigma}_{\phi^*}^2$ is the complete-data asymptotic variance estimate and λ is the ratio $(\phi^{(p+1)} - \phi^*) / (\phi^{(p)} - \phi^*)$ for large p . Of course it is often the case in multi-parameter problems that preliminary estimates are used for likelihood ratio testing, and corresponding estimates of the variances are not necessary.

Mr Orchard suggests that further details relating to specific examples, both those we mentioned and others, are very much worth pursuing. We of course agree. Mr Tornsey, Dr E. A. Thompson and Professor Turnbull all indicate directions for such work. We are continuing to work actively along these lines. Laird (1975) studies variance estimation for random parameters in log-linear models for contingency tables. Laird (1976) discusses non-parametric estimation of a univariate distribution where observations have errors whose distribution is specified in parametric form. Dempster and Monajemi (1976) present further details of variance components estimation from the EM standpoint, and we believe they reply to many of the issues raised by Professor Searle. Papers involving iteratively reweighted least squares (DLR), factor analysis (DR) and rounding error in regression (DR) are nearing completion.

Several discussants question the usefulness of the general definition of the EM given in equation (2.17) and successive lines. The essence of the question is that an algorithm is undefined unless the specific computational steps are enumerated in such a way that they can be numerically implemented. Professor Hartley points out that the E-step is most easily implemented when the distributions are discrete. In continuous exponential families cases, there are sometimes simple analytic expressions for the necessary expectations and for $a(\phi)$, but, in general, specification of the E-step for continuous distributions requires numerical integration. Note that both $E\{t | \phi^{(p)}, \mathbf{y}\}$ and $a(\phi)$ are defined as integrals, but over different spaces. Dr Nelder is generally right to assert that unless the parametric space Ω is discrete, $Q(\phi | \phi^{(p)})$ can be evaluated numerically only selectively at points on a grid, and similarly we accept Beale's remark, "... division of the method into separate

E and M steps is quite impractical". In general, the computational task of passing from $\phi^{(p)}$ to $\phi^{(p+1)}$ will itself be iterative and will involve a sequence of steps $\phi^{(p,r)}$ for $r = 1, 2, \dots$. An important problem is to minimize the number of points ϕ where $Q(\phi | \phi^{(p)})$ is computed during the inner loop. That is, efficient mixing of E- and M-steps is required.

It thus appears that the strict separation of E- and M-steps is numerically feasible only when the M-step is rather easily computable given the expectations of a finite number of sufficient statistics. We used the E-step in our general formulation as a separate entity chiefly because of the statistical principle that it expresses: the idea is to maximize the estimated complete-data log-likelihood as though it were the actual log-likelihood.

Some of the comments advising against uncritical use of EM algorithms bear not on the existence of better algorithms but rather on the question of whether maximum likelihood is a good method in specific applications. Although we did not discuss good and bad statistics in our paper, we certainly share the concern that the availability of easy computer methods may lead to bad practice of statistics. Statisticians who use likelihood methods have a responsibility to assess the robustness of their conclusions against failures in the specific parametric models adopted. Even accepting the parametric forms, there are reasons to be suspicious of routine use of maximum likelihood, especially when a large number of parameters is involved. To take an example raised by Beale, suppose a variable Y is to be predicted from a substantial number of independent variables $X_1, X_2, \dots, X_p$ when it is assumed that all the variables are jointly normally distributed. If the ratio of sample size n to p is not large, then maximum likelihood gives an estimate of the residual variance of Y that is badly biased. With complete data, there is a standard correction, whereby the residual sum of squares is divided by $n - p - 1$ instead of n but, as far as we know, there is no such standard correction available when a substantial amount of data in the X matrix is missing. One important logical approach to improving maximum likelihood in such situations is to model the parameters, that is, to regard the parameters as randomly drawn from a distribution which itself has relatively few parameters. Factor analysis seems to us to be especially in need of treatment of this kind. We share with Mr Orchard interest in practical applications of these more Bayesian approaches.

Some of the purely numerical problems of EM are symptomatic of difficult statistical problems. We enjoyed Murray's simple example of multiple maxima, but more important is his remark that multiple maxima occur frequently in practice. The phenomenon is well known in the area of estimating mixtures. Multiple maxima suggest that the familiar quadratic approximation to log-likelihood may not be adequate, so that the shape of the likelihood surface needs investigating, and we should not accept as adequate a few standard summary statistics based on derivatives at the maxima.

We did not intend to suggest that the mathematical results of Baum *et al.* (1970) are of limited mathematical generality, but only that the wide range of application of these results to statistical problems was not recognized in their article.

We wish to reiterate our debt to Professor Hartley, who is the originator of many of the ideas reviewed in our paper. We think that we have brought the techniques into better focus, clarified the mathematics, and shown that the range of important examples is substantially greater than was previously thought. As Professor Efron points out, R. A. Fisher long ago used the basic first derivative relation of Sundberg (1974) in the special context of inefficient statistics, but without the specific application to incomplete data problems as discussed by Hartley.

We are less happy with some of Professor Hartley's technical comments. For example, he asserts that the treatment of truncated distributions in his 1958 paper does not fall within our framework, but in fact our paper contains a specific technical contribution showing how the case of truncation does fall within our framework. This work begins in the fifth paragraph of Section 4.2, and treats a general form of truncation including Hartley's example as a special case.

We believe that our references to "ridges" are clear, referring in all cases to ridges in the actual likelihood $g(y | \phi)$. Obviously when there are ridges in $f(x | \phi)$ the parameters in the complete-data model are not identifiable, and the M-step is essentially undefined. We regard this case as uninteresting. Hartley and Hocking (1971) assume there is no ridge in $g(y | \phi)$, and we point out that convergence generally obtains even when this condition fails. Our best example is the factor analysis model. Mr Healy specifically, and Dr Ord implicitly, ask us whether the nonuniqueness of the specific basis chosen for factors interferes with convergence of the algorithm. The answer is simply "no", because the steps of the algorithm are defined in a coordinate-free way. Our

convergence proofs merely generalize what obviously holds in the special case of factor analysis. Theorems 2 and 3 of Hartley and Hocking (1971) prove convergence of the EM algorithm under conditions which are much more restrictive than our conditions. As Mr Beale remarks, our Theorem 1 together with an assumption of bounded likelihood obviously implies the existence of a convergent subsequence. Our further theorems specify nontrivial conditions which are often verifiable and which rule out multiple limit points.

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